

Gut bacteria alleviate smoking-related NASH by degrading gut nicotine

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Tobacco smoking is positively correlated with non-alcoholic fatty liver disease (NAFLD)^{1–5}, but the underlying mechanism for this association is unclear. Here we report that nicotine accumulates in the intestine during tobacco smoking and activates intestinal AMPK α . We identify the gut bacterium *Bacteroides xylanisolvens* as an effective nicotine degrader. Colonization of *B. xylanisolvens* reduces intestinal nicotine concentrations in nicotine-exposed mice, and it improves nicotine-exacerbated NAFLD progression. Mechanistically, AMPK α promotes the phosphorylation of sphingomyelin phosphodiesterase 3 (SMPD3), stabilizing the latter and therefore increasing intestinal ceramide formation, which contributes to NAFLD progression to non-alcoholic steatohepatitis (NASH). Our results establish a role for intestinal nicotine accumulation in NAFLD progression and reveal an endogenous bacterium in the human intestine with the ability to metabolize nicotine. These findings suggest a possible route to reduce tobacco smoking-exacerbated NAFLD progression.

As one of the most common unhealthy behaviours among humans, the World Health Organization lists tobacco smoking as the leading cause of preventable death in the world^{6,7}. Smoking cessation is the most effective way to extend life expectancy, but it is not easily achieved owing to the powerful addictive property and withdrawal symptoms of nicotine⁸. Nicotine degradation has recently been considered as a strategy for blocking nicotine-induced pathology^{9,10}. NAFLD encompasses a broad spectrum of pathologies that usually progress from simple steatosis (defined as non-alcoholic fatty liver (NAFL)) to NASH and then, in some cases, to cirrhosis and hepatocellular carcinoma. As metabolites produced from the intestine have a critical role in the progression of NAFLD through the gut–liver axis¹¹, establishing a relationship between intestinal nicotine accumulation, intestine-derived metabolites and NAFLD progression could shed fresh light on tobacco-smoking-related liver diseases. Here we report that nicotine accumulates in the intestine during tobacco smoking and accelerates the progression of NAFLD, but that it can be degraded effectively by the human symbiont *B. xylanisolvens*. These findings address the pathological effects of intestinal nicotine accumulation, and identify an endogenous nicotine-degrading gut bacterium that is potentially valuable for the prevention and treatment of NASH in smokers.

Gut bacteria degrade ileal nicotine

High levels of nicotine can be detected in human saliva and gastric juice during tobacco smoking or nicotine patch application, indicating its accumulation in the digestive tract¹², in which it may act pathologically. To confirm whether nicotine accumulates in the intestine during tobacco smoking, we collected terminal ileum mucosa biopsy, serum and stool samples from 30 non-smokers and 30 smokers. High levels of nicotine were detected in the ileal mucosa tissue of smokers, suggesting that nicotine accumulates in the intestine during tobacco smoking (Fig. 1a). We further assessed nicotine levels in the lungs, ileal contents, ileal tissue, brain, liver, epididymal white adipose tissue and serum in specific-pathogen-free (SPF) mice given nicotine through three previously described methods of administration (that is, smoke exposure, oral and subcutaneous)^{13–16}. Ileal nicotine accumulation was consistently observed in the above mouse models (Fig. 1b,c and Extended Data Fig. 1a). These results indicate that nicotine levels during smoking accumulate in the intestine to a relatively large degree, which may be of pathophysiological significance.

In the nicotine drinking water delivery model, the nicotine concentrations in both the ileal content and ileal tissue in germ-free (GF) mice

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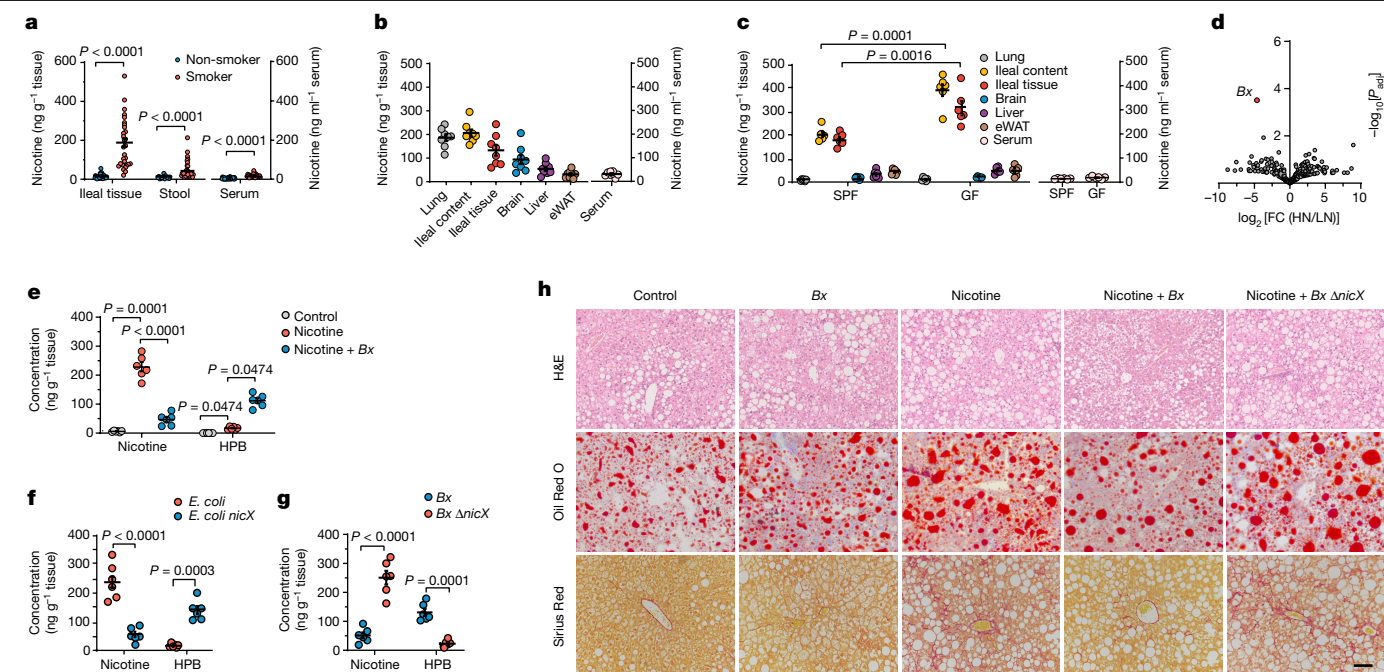


Fig. 1 | Identification of intestinal nicotine accumulation and gut-bacteria-derived intestinal nicotine degradation. **a**, Quantification of the nicotine concentrations in human terminal ileum mucosa biopsy, serum and stool samples collected from smokers ($n = 30$) and non-smokers ($n = 30$). For the assessment of ileal nicotine accumulation, the indicated mouse models in **b**, **c** and **e–g** were treated for two weeks. **b**, Nicotine concentrations in lung, ileum content, ileum tissue, brain, liver, epididymal white adipose tissue (eWAT) and serum samples in the SPF mouse smoke exposure model. $n = 8$ mice per group. **c**, Tissue nicotine concentrations of SPF and GF mice in the nicotine drinking model. $n = 6$ mice per group. **d**, Human stool metagenomic sequencing data in the HN and LN groups. The adjusted P value was calculated using the moderated Student's t -test followed by the Benjamini–Hochberg procedure via the false-discovery rate (FDR). FC, fold change. **e**, Nicotine and HBP concentration in ileal tissue of control-, nicotine (in drinking water)- and

nicotine + *B. xylanisolvens* (*Bx*)-treated SPF mice, separately. $n = 6$ mice per group. **f,g**, The nicotine and HBP concentration in the ileal tissue of SPF mice treated with *E. coli* or *nicX*-knockin *E. coli* (**f**) and SPF mice treated with *B. xylanisolvens* or *nicX*-knockout *B. xylanisolvens* (**g**) under nicotine drinking water treatment. $n = 6$ mice per group. HFHCD-fed SPF mice were treated with PBS, *B. xylanisolvens* colonization, nicotine water, nicotine water + *B. xylanisolvens* colonization and nicotine water + *nicX*-knockout *B. xylanisolvens* colonization for 20 weeks. $n = 6$ mice per group. **h**, Representative H&E (top), Oil Red O (middle) and Sirius Red (bottom) staining of liver sections. $n = 3$ mice per group, 3 images per mouse. Scale bar, 100 μ m. For **a–g**, data are mean $\pm$ s.e.m. Statistical analysis was performed using two-tailed Mann–Whitney U -tests (**a**); two-tailed Student's t -tests (**c**, **f** and **g**); one-way analysis of variance (ANOVA) with Dunnett's T3 post hoc test (**e**, nicotine); and Kruskal–Wallis test with Dunn's test (**e**, HBP).

were higher than those in SPF mice, suggesting that endogenous gut bacteria contribute to nicotine degradation (Fig. 1c). To identify the specific endogenous nicotine-degrading gut microbiota in humans, we screened for potential strains carrying known nicotine catabolic genes from public metagenome databases. A homologous sequence of *nicA*—one of the known nicotine-degrading enzymes^{17,18}—was found in several strains, suggesting that such strains have the potential to degrade nicotine (Extended Data Fig. 1b). Stool samples from 30 smokers were analysed using a metagenomics study to identify the most correlative species with reduced ileal nicotine concentrations. The participants were defined as either low-nicotine (LN) or high-nicotine (HN), according to the ileal nicotine levels (Extended Data Fig. 1c). Notably, there was no significant difference between the LN and HN groups regarding smoking history, including years of smoking and number of cigarettes per day (Supplementary Tables 1 and 2). No differences in the α - and β -diversities were found between the two groups; however, several strains showed different abundances between the LN and HN groups (Fig. 1d and Extended Data Fig. 1d–g). On the basis of metagenome databases and metagenomics, we found that *B. xylanisolvens* potentially contained a nicotine catabolic gene and was identified as the main driver associated with the separation between the LN and HN groups (Fig. 1d and Extended Data Fig. 1h,i).

Supplementation of nicotine in the culture medium had no influence on the growth rate of *B. xylanisolvens* (Extended Data Fig. 2a); however, *B. xylanisolvens* efficiently degraded the nicotine that was added into the medium (Extended Data Fig. 2b). To examine the degraded

metabolites of nicotine in *B. xylanisolvens*, we performed fermentation of *B. xylanisolvens* and isolated the nicotine metabolite using chromatography. The metabolite was further characterized using nuclear magnetic resonance (NMR), and was identified as 4-hydroxy-1-(3-pyridyl)-1-butanone (HPB; Extended Data Fig. 2c), which could be detected in the culture medium of *B. xylanisolvens* (Extended Data Fig. 2d). To verify the nicotine-degrading properties of *B. xylanisolvens* in vivo, we colonized this bacterium into SPF mice that were administered nicotine through the three delivery strategies described above and found that *B. xylanisolvens* could degrade nicotine in vivo, accompanied by an increase in HPB (Fig. 1e and Extended Data Fig. 2e,f).

Next, we performed whole-genome sequencing of *B. xylanisolvens* and examined the biosynthetic gene responsible for the catabolism of nicotine in *B. xylanisolvens*. Among the potential nicotine-catabolic proteins in *B. xylanisolvens* encoded by these sequenced genes, NicX (encoded by *nicX*) has a similar predicted core structure to the computational model of the known nicotine-degrading enzyme NicA (Extended Data Fig. 2g). NicX degraded nicotine with a $V_{\max}$ of $3.83 \pm 0.13 \mu\text{mol min}^{-1}$ per μg protein in vitro (Extended Data Fig. 2h). To further verify the nicotine-degrading ability of NicX, *nicX* was introduced into *Escherichia coli* or knocked out in *B. xylanisolvens* (Extended Data Fig. 2i). Compared with wild-type (WT) *E. coli*, *nicX*-knockin *E. coli* gained the ability to degrade nicotine and produce HPB in vitro and in vivo (Fig. 1f and Extended Data Fig. 2j). Moreover, *nicX* deficiency in *B. xylanisolvens* resulted in the loss of its nicotine-degradation ability (Fig. 1g and Extended Data Fig. 2k). No significant change was found

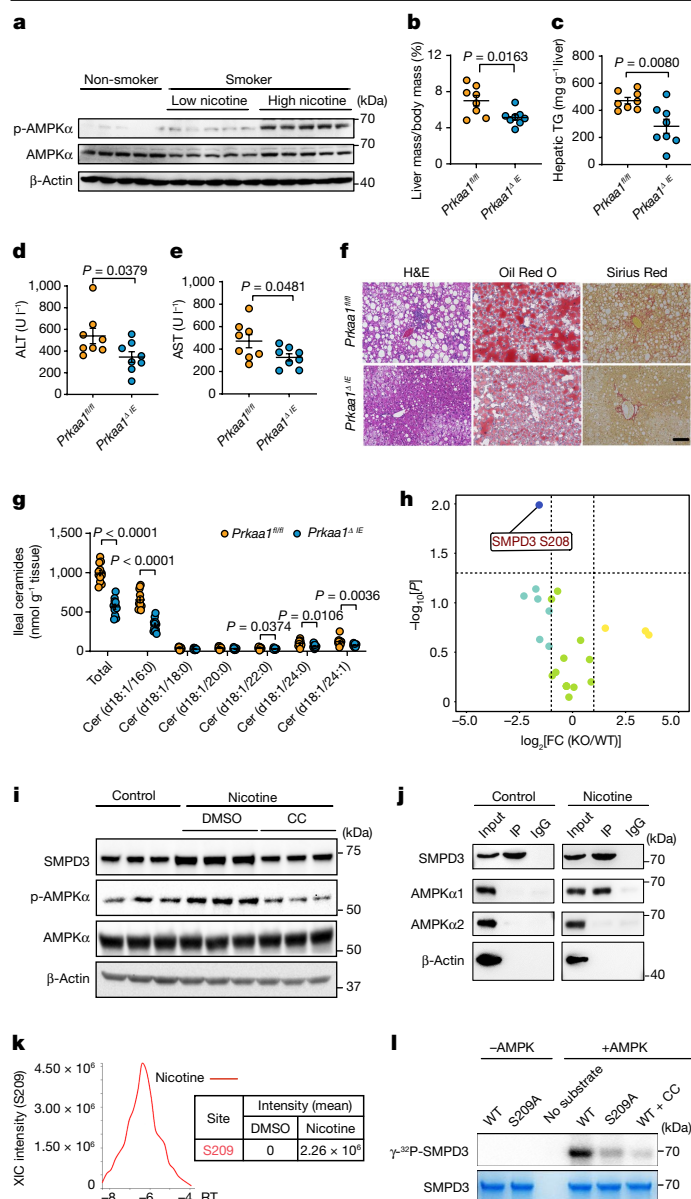


Fig. 2 | Nicotine-induced activation of the intestinal AMPKα-SMPD3 axis in NAFLD progression. **a**, Representative western blot of ileal AMPK signalling in non-smokers ($n = 30$) and smokers with low-nicotine ($n = 14$) or high-nicotine ($n = 16$) contents in terminal ileum mucosa biopsy samples. **b–f**, *Prkaa1^{fl/fl}* and *Prkaa1^{ΔIE}* mice (SPF) were administered a HFHCD and nicotine water for 20 weeks. $n = 8$ mice per group. **b**, Ratios of liver mass to body mass. **c**, Liver triglyceride (TG) content. **d, e**, Serum ALT (**d**) and AST (**e**) levels. **f**, Representative H&E (left), Oil Red O (middle) and Sirius Red (right) staining of liver sections. $n = 4$ mice per group, 3 images per mouse. Scale bar, 100 μ m. **g**, Quantification of ileal ceramide (cer) profiles between *Prkaa1^{fl/fl}* (WT, $n = 12$) and *Prkaa1^{ΔIE}* (KO, $n = 11$) mice administered a HFHCD and nicotine water for 20 weeks. Different types of ceramide are designated by a slash followed by the carbon chain length and number of double bonds in the fatty acid. **h**, Volcano map showing all phosphorylated sites of ceramide-metabolism-related proteins identified from phosphorylated proteomics analysis of the ileal epithelia of *Prkaa1^{fl/fl}* (WT) and *Prkaa1^{ΔIE}* (KO) mice administered a HFHCD and nicotine water for 20 weeks. $n = 5$ mice per group. P values were calculated using two-sided Student's t -tests. **i**, Assessment of AMPK activation and SMPD3 protein levels in ileal organoids treated with control, nicotine or nicotine plus compound C (CC, AMPK inhibitor) for 12 h. **j**, Co-immunoprecipitation analysis of AMPKα1 with SMPD3. Constructs encoding Flag-tagged SMPD3 were transfected into SW480 cells, and then treated with nicotine for 4 h. **k**, MS analysis of the phosphorylation intensity at Ser209 of SMPD3 with or without nicotine treatment for 24 h. XIC, extracted ion chromatogram; RT, retention time. **l**, In vitro phosphorylation of WT or S209A mutant SMPD3 (86–655) by AMPK ($\alpha 1$ – $\beta 1$ – $\gamma 2$) with or without compound C treatment. For **b–e** and **g**, data are mean $\pm$ s.e.m. Statistical analysis was performed using two-tailed Student's t -tests (**b**, **c** and **e**) and two-tailed Mann–Whitney U -tests (**d**, **g**). The experiments in **i–l** were performed three times independently.

in the growth curves of the WT and *nicX*-knockout *B. xylanisolvens* (Extended Data Fig. 2l). In summary, the results show that nicotine accumulates in the intestine during various routes of nicotine administration, and *B. xylanisolvens* has the ability to degrade nicotine in the presence of NicX.

Nicotine degradation delays NAFLD progression

Oral nicotine administration through drinking water is a widely used model to study nicotine-associated chronic metabolic diseases^{19,20}. To determine the role of intestinal nicotine accumulation and the effect of its degradation by *B. xylanisolvens* in NAFLD progression, SPF mice treated with PBS (control), *B. xylanisolvens*, nicotine water, *B. xylanisolvens* plus nicotine water or *nicX*-knockout *B. xylanisolvens* plus nicotine water were administered a high-fructose and high-cholesterol diet (HFHCD) for 20 weeks. WT and *nicX*-knockout *B. xylanisolvens* were both successfully colonized in HFHCD-fed SPF mice (Extended Data Fig. 3a). However, nicotine degradation was observed only in the group colonized with WT *B. xylanisolvens* (Extended Data Fig. 3b). No obvious change in body mass was observed among the groups (Extended Data Fig. 3c, d). The liver masses, serum levels of alanine aminotransferase

(ALT) and aspartate aminotransferase (AST), and hepatic triglyceride levels of nicotine-treated mice were higher compared with those of control-treated SPF mice (Extended Data Fig. 3e–i). No significant differences in the serum triglyceride, the hepatic and serum cholesterol ester and serum non-esterified fatty acid levels were found between the two groups (Extended Data Fig. 3j–m). On the basis of haematoxylin and eosin (H&E), Oil Red O and Sirius Red staining, nicotine supplementation in the water accelerated NAFLD progression under HFHCD feeding conditions, and this acceleration was accompanied by more severe hepatic steatosis, inflammation and fibrosis (Fig. 1h and Extended Data Fig. 3n–r). Furthermore, increases in the relative expression of mRNAs involved in hepatic lipid metabolism, proinflammatory cytokine production and collagen synthesis were induced by nicotine-supplemented water treatment (Extended Data Fig. 3s–u), whereas these NASH indicators were ameliorated by colonization with WT *B. xylanisolvens* but not with *nicX*-deficient *B. xylanisolvens* (Fig. 1h and Extended Data Fig. 3c–u). Together, these data suggest that the colonization of nicotine-degrading bacteria could reduce nicotine-induced NASH, and this effect is dependent on the expression of NicX.

Nicotine activates intestinal AMPKα1

Consistent with previous studies²¹, we found that nicotine treatment induced the phosphorylation of AMPKα at Thr172 in ileal organoids in a dose-dependent manner (Extended Data Fig. 4a, b). Furthermore, ileal AMPKα was activated by tobacco smoking in the terminal ileum, and the strength of the effect was positively related to the level of nicotine in the tissue of smokers (Fig. 2a). Moreover, phosphorylated AMPKα (p-AMPKα) in the ileum was increased in all three models of nicotine-exposed SPF mice (Extended Data Fig. 4c–e).

There are two isoforms of AMPKα, including AMPKα1 and AMPKα2 (ref. 22). We generated intestinal-epithelium-specific AMPKα1 (*Prkaa1^{ΔIE}*) and AMPKα2 (*Prkaa2^{ΔIE}*) knockout mouse lines (Extended Data Fig. 4f). Nicotine completely lost its ability to activate AMPKα in the absence of intestinal AMPKα1, whereas the nicotine-induced increase in p-AMPKα still occurred in the absence of intestinal AMPKα2, suggesting that

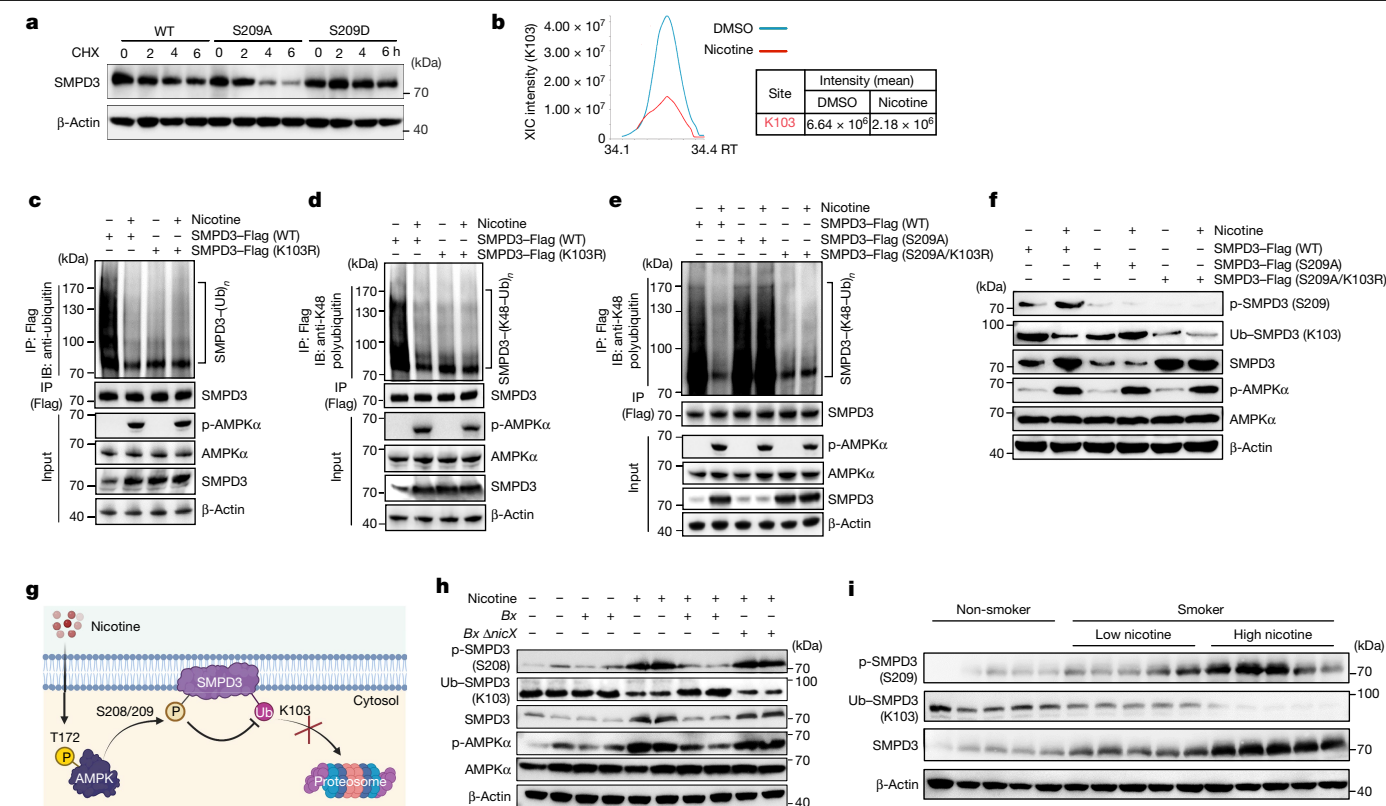


Fig. 3 | Phosphorylated SMPD3 is more stable due to reduced ubiquitination-mediated degradation. **a**, Western blot analysis showing the stability of SMPD3 after the administration of nicotine and cycloheximide (CHX, translation inhibitor) in SW480 cells transfected with WT or mutant (S209A or S209D) SMPD3. **b**, MS analysis of the ubiquitination intensity at Lys103 of SMPD3 with or without nicotine treatment. **c**, Total ubiquitination of SMPD3 in SW480 cells transfected with WT or K103R SMPD3-Flag with or without nicotine treatment. **d**, Lys48 ubiquitination of SMPD3 in SW480 cells transfected with WT or K103R SMPD3-Flag with or without nicotine treatment. **e**, Lys48 ubiquitination of SMPD3 in SW480 cells transfected with WT, S209A or S209A/K103R SMPD3-Flag without or with nicotine treatment. **f**, Western blot analysis showed phosphorylation levels, ubiquitination levels and protein levels of SMPD3 in SW480 cells transfected with three SMPD3-Flag plasmids (WT, S209A, S209A/K103R) and treated with or

without nicotine. **g**, Diagram of the SMPD3 protein phosphorylation and ubiquitination under normal and nicotine-stimulated conditions. This diagram was created using BioRender. **h**, Representative ileal SMPD3 protein phosphorylation and ubiquitination levels in HFHCD-fed SPF mice treated with PBS, *B. xylanisolvens* colonization, nicotine water, nicotine water + *B. xylanisolvens* colonization, and nicotine water + *nicX*-knockout *B. xylanisolvens* colonization for 20 weeks. *n* = 4 mice per group. **i**, Representative western blot of ileal SMPD3 phosphorylation, ubiquitination and protein levels in non-smokers and smokers with low nicotine or high nicotine concentrations in terminal ileum mucosa biopsy samples. Experiments were performed with 30 non-smokers and 30 smokers (14 with low nicotine levels and 16 with high nicotine levels). For **b–f**, nicotine (1 μg ml⁻¹) treatment was administered for 24 h. The experiments in **a–f** were performed three times independently.

AMPKα1 is the main target of nicotine in the intestine (Extended Data Fig. 4f). *B. xylanisolvens* colonization inhibited nicotine-induced AMPKα phosphorylation in the ileum (Extended Data Fig. 4g). Similar to *B. xylanisolvens*, *nicX*-knockin *E. coli* had the ability to suppress the nicotine-induced phosphorylation of ileal AMPKα in vivo (Extended Data Fig. 4h). However, the *B. xylanisolvens*-mediated inhibitory effects were not observed in the absence of NicX (Extended Data Fig. 4i). Thus, *B. xylanisolvens* mitigates nicotine-induced activation of ileal p-AMPKα in the presence of NicX, which is supported by the observation that the nicotine catabolite HPB could not activate ileal p-AMPKα (Extended Data Fig. 4j).

Regarding the role of intestinal AMPKα1 in nicotine-accelerated NAFLD progression, we found that the loss of intestinal epithelial AMPKα1 contributes to improvements in hepatic steatosis, inflammation and fibrosis in the nicotine-accelerated NASH mouse model (Fig. 2b–f and Extended Data Fig. 5a–m).

p-AMPKα phosphorylates SMPD3 at Ser208/209

Lipids are considered to have an important role in NAFLD progression, as they are a key mediator of the crosstalk between the intestine and liver in this pathology²³. We conducted a targeted lipidomic analysis of

the ileum and identified ceramides as the primary metabolites leading to clustering-based differentiation between *Prkaa1*-knockout (KO) and control (WT) intestinal epithelia with nicotine treatment (Extended Data Fig. 5n,o). Ileal ceramide profiles were lower in the absence of intestinal AMPKα1 in the nicotine-drinking model (Fig. 2g). On the basis of phosphoproteome analysis of the ileal epithelium from the nicotine-drinking model (Extended Data Fig. 5p), phosphorylation at Ser208 of SMPD3—the enzyme that mediates sphingomyelin conversion to ceramide—was significantly decreased by intestinal epithelium AMPKα1 loss (Fig. 2h and Extended Data Fig. 5q).

In ileal organoids, nicotine treatment did not alter *Smpd3* mRNA levels (Extended Data Fig. 6a). However, either treatment with an AMPK kinase activity inhibitor or expression of a kinase-domain mutant could mitigate the nicotine-induced increase of SMPD3 protein levels (Fig. 2i and Extended Data Fig. 6b), suggesting that the effect of nicotine on SMPD3 is post-translational. Indeed, we found that, under nicotine treatment, SMPD3 bound to AMPKα1 (Fig. 2j). A GPS 2.0 (ref. ²⁴) analysis predicted that Ser208 (mouse) is the most likely phosphorylation site by AMPK (Extended Data Fig. 6c). Peptides including Ser208 (Ser209 in humans) in SMPD3 satisfied the phosphorylation substrate motif of AMPK and were highly evolutionarily conserved (Extended Data Fig. 6d). The stability of SMPD3 is reportedly regulated by the phosphorylation

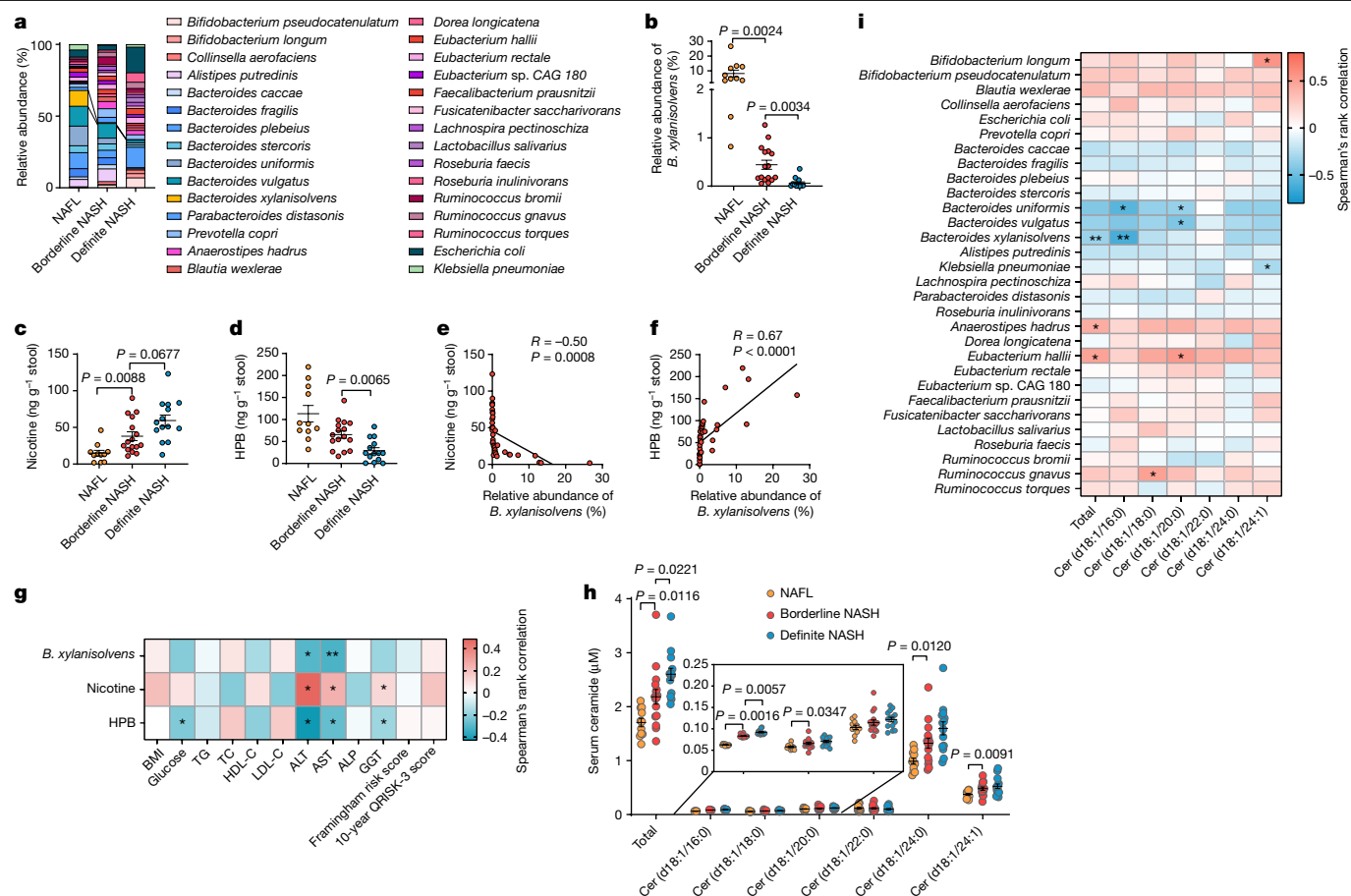


Fig. 4 *B. xylanisolvens*-mediated nicotine degradation is negatively correlated with clinical NAFLD progression. **a–i**, Analysis of 41 smokers with NAFLD, including NAFL ($n = 11$), borderline NASH ($n = 16$) and definite NASH ($n = 14$). **a**, Bacterial taxonomic profiling of the gut microbiota from different NAFLD stages processed at the species level. **b**, Relative abundances of *B. xylanisolvens* associated with different NAFLD stages in smokers with NAFLD. **c,d**, Concentrations of faecal nicotine (**c**) and HPB (**d**) at different NAFLD stages in smokers with NAFLD. **e,f**, Correlative analysis of *B. xylanisolvens* with faecal nicotine (**e**) and HPB (**f**). Correlations between variables were assessed by linear

regression analysis. Linear correlation index R and P values were calculated. **g**, The correlation between *B. xylanisolvens* abundance, faecal nicotine and HPB levels and metabolic indicators. Correlation analyses were performed using Spearman's rank tests; $*P < 0.05$, $**P < 0.01$. BMI, body mass index. **h**, Quantification of serum ceramides in different NAFLD stages. **i**, The correlation between gut bacteria (top 30) and serum ceramides levels. Correlation analyses were performed using Spearman's rank tests; $*P < 0.05$, $**P < 0.01$. For **b–d** and **h**, data are mean $\pm$ s.e.m. Statistical analysis was performed using Kruskal–Wallis test with Dunn's test (**b, c** and **h**) and one-way ANOVA with Dunnett's T3 post hoc test (**d**).

of five conserved serine sites²⁵. Consistently, increased Ser209 phosphorylation was detected in the nicotine-treated group (Fig. 2k and Extended Data Fig. 6e). SMPD3 (WT) could be directly phosphorylated by AMPK α 1, and this phosphorylation was almost abolished when Ser209 was mutated or when an AMPK inhibitor was administered (Fig. 2l). To confirm whether phosphorylation at Ser209 influences SMPD3 protein stability, SW480 cells were transfected with WT, S209A (lost phosphorylation mutant) and S209D (sustained phosphorylation mutant) SMPD3 and treated with nicotine and cycloheximide (CHX, a protein translation inhibitor). We found that Ser209 phosphorylation protected SMPD3 protein from degradation (Fig. 3a). In isolated organoids from the nicotine-drinking model, faster degradation of SMPD3 was observed in organoids from mice lacking intestinal epithelium AMPK α 1 (Extended Data Fig. 6f).

Furthermore, SMPD3 was notably more deubiquitinated at Lys103 in nicotine-treated SW480 cells compared with in DMSO-treated cells (Fig. 3b and Extended Data Fig. 6g). Using immunoprecipitation, we found that the ubiquitinated SMPD3(Lys103) contained primarily the classical Lys48-linked Ub chain (Lys48–Ub) instead of Lys63–Ub, which was decreased by nicotine treatment (Fig. 3c,d and Extended Data Fig. 6h). S209A mutation reversed the nicotine-induced decrease in Lys48–Ub of SMPD3, which was further mitigated by the K103R

mutation (Fig. 3e and Extended Data Fig. 6i). Moreover, we developed anti-p-SMPD3 (S209) and anti-Ub–SMPD3(Lys103) antibodies (Extended Data Fig. 6j,k), and confirmed that the nicotine-induced increases in total SMPD3 protein resulted from elevated SMPD3 phosphorylation and subsequently reduced ubiquitination (Fig. 3f). These findings suggest that nicotine-activated AMPK α phosphorylates SMPD3 at Ser209, which in turn inhibits Lys103 ubiquitination on SMPD3 and the subsequent Lys48–Ub degradation pathway, thereby increasing SMPD3 protein stability and its protein level (Fig. 3g). We further detected nicotine-induced SMPD3 phosphorylation at Ser208, accompanied by a decrease in SMPD3 ubiquitination at Lys103 in the ileal tissues of SPF mice, which was mitigated by *B. xylanisolvens* colonization in the presence of NicX (Fig. 3h). Phosphorylated and total SMPD3 protein levels were also higher in the ileum of smokers compared with non-smokers, which was related to ileal nicotine content; however, ubiquitinated SMPD3 protein levels were decreased sequentially in the non-smoker, LN and HN groups (Fig. 3i).

The p-AMPK α –SMPD3–ceramide axis and NASH

SMPD3 is considered to be an accelerator of pathology in metabolic diseases²⁶. GW4869, a SMPD3 inhibitor, was administered to organoids

to suppress SMPD3 activity (Extended Data Fig. 7a). The ceramide levels in organoids and their supernatants under nicotine treatment were decreased by GW4869 (Extended Data Fig. 7b,c). Lentivirus-expressed SMPD3 was applied to restore the expression of SMPD3 in *Prkaa1*-knockout ileal organoids from the nicotine-drinking model (Extended Data Fig. 7d). As expected, SMPD3 overexpression reversed the reduction in ceramide levels resulting from AMPK α 1 deficiency in ileal organoids (Extended Data Fig. 7e,f). In the nicotine-accelerated NASH mouse model, *B. xylanisolvans*-colonized SPF mice showed a reduction in ileal ceramides in the presence of NicX (Extended Data Fig. 7g), which resulted from the inhibition of the intestinal p-AMPK α -SMPD3 axis.

To confirm whether modulation of the ceramide profiles accounts for the NASH improvement derived from intestinal epithelium AMPK α 1 deficiency in the nicotine-accelerated NASH mouse model, we administered ceramide (d18:1/16:0) by daily intraperitoneal (i.p.) injection (Extended Data Fig. 8a). The protective effects on hepatic steatosis, inflammation and fibrosis resulting from the genetic disruption of intestinal epithelium AMPK α 1 were reversed by ceramide (d18:1/16:0) administration (Extended Data Fig. 8b–s). Moreover, oral GW4869 delivery decreased the ileal levels of ceramides, and the decrease was accompanied by lower NAFLD severity in the nicotine-drinking model (Extended Data Fig. 9a–s). In conclusion, nicotine-induced activation of the intestinal AMPK α -SMPD3 axis potentiates NAFLD progression by increasing intestinal ceramide production, and thus SMPD3 suppression is a potential strategy for relieving hepatic steatosis, inflammation and fibrosis.

B. xylanisolvans and clinical NAFLD

To elucidate the correlation between our findings and human NAFLD progression, we collected data from 83 patients with biopsy-proven NAFLD, who were divided into smokers ($n = 41$) and non-smokers ($n = 42$) (Supplementary Tables 3–6). Among the smokers, there was no significant difference between NAFL, borderline NASH and definite NASH with respect to tobacco smoking history, including years of tobacco smoking and number of cigarettes per day (Supplementary Tables 3 and 4). However, we found that *B. xylanisolvans* levels were negatively correlated with NAFLD severity (that is, from NAFL to borderline NASH to definite NASH), as indicated by the NAFLD-activity score (NAS) (Fig. 4a,b and Extended Data Fig. 10a–c). As a result, the faecal nicotine levels were step-wise increased with the progression of NAFLD, and the nicotine-degradation product HPB in faeces was accordingly decreased along with NAFLD progression (Fig. 4c,d). In smokers, *B. xylanisolvans* abundance was negatively correlated with faecal nicotine content, but positively correlated with HPB levels (Fig. 4e,f).

In terms of metabolic indicators, the presence of *B. xylanisolvans* and HPB in the faeces was found to be inversely correlated to serum ALT and AST levels, whereas nicotine concentrations were positively correlated (Fig. 4g). These results suggest that *B. xylanisolvans*-mediated nicotine degradation potentially safeguards liver function in individuals who smoke. The serum ceramide levels, especially ceramide (d18:1/16:0), which is mainly derived from the intestine, were significantly higher in individuals with NASH compared with those with NAFL (Fig. 4h), and it was negatively correlated with the abundance of *B. xylanisolvans* (Fig. 4i).

However, we did not observe a correlation between *B. xylanisolvans* abundance and NAFLD severity in non-smokers (Extended Data Fig. 10d–j). In conclusion, these data establish intestinal nicotine accumulation as a risk factor for clinical NAFLD progression in smokers, while also suggesting that *B. xylanisolvans*, with its nicotine-degrading activities, could exhibit protective effects by decreasing the progression of NAFLD (Extended Data Fig. 10k).

Discussion

In patients who smoke and in three nicotine-exposed mouse models, high concentrations of intestinal nicotine were detected using a method

based on liquid chromatography coupled with mass spectrometry. Thus, the distribution of nicotine in the intestine exists irrespective of the route of delivery. Intraperitoneal nicotine supplementation in mice induced nicotine accumulation in the stool¹⁶, and previous literature using nicotine skin patches in humans also found an obvious accumulation of nicotine in the saliva and gastric juice²⁷, although the mechanisms for this accumulation needs to be addressed in future studies.

Although some natural nicotine-degrading bacteria have been proposed^{28,29}—suggesting a bacteria-based approach to lower nicotine levels—the nicotine-degrading properties of the gut microbiota in mammals have yet to be confirmed. Our findings raised the potential of gut microbial nicotine degradation. *B. xylanisolvans* has been verified to be safe for in vivo use, and the European Union has authorized the inclusion of heat-treated milk products fermented with *B. xylanisolvans* DSM 23964 as a food³⁰. Moreover, HPB, a nicotine-derived catabolic metabolite, could potentially be used as a biomarker to assess the degree of gut-microbial-mediated nicotine catabolism in patients. Even so, more research is needed to fully understand the gut microbial nicotine-degrading ability.

The identification of the AMPK substrate SMPD3 provides the first link from nicotine-activated AMPK to ceramide metabolism, and this pathway may also explain the promotion of ceramide levels by cigarette smoke in serum and other tissues, such as lungs, liver and abdominal aorta^{31,32}. In addition to a high-fat diet and a high-cholesterol diet, here we show that nicotine intake is another important inducer of intestine-derived ceramide production. On the basis of our findings, this study indicates that interventions with either microbial nicotine degraders or other targets based on the intestinal AMPK α -SMPD3-ceramide axis show translational potential for the treatment of nicotine-associated NASH.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-022-05299-4>.

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Methods

Human participants

Three cohorts were included in this study. Cohort 1 was approved by the Ethical Committee of the First Affiliated Hospital, Zhejiang University, School of Medicine (2021(875)). All of the participants provided written informed consent. Cohort 2 and cohort 3 were approved by the Ethical Committee of the First Affiliated Hospital of Wenzhou Medical University (2016-246) and informed written consent was obtained from all participants.

For cohort 1, terminal ileum mucosa biopsy, serum and stool samples were collected from 30 non-smokers and 30 smokers. The characteristics of the participants are listed in Supplementary Tables 1 and 2. All of the participants were diagnosed with colon polyps. During the polypectomy under a colonoscopy, we obtained trace amounts of terminal ileal mucosa tissue with no obvious abnormalities.

For cohort 2, serum and stool samples were collected from 41 smokers diagnosed with biopsy-confirmed NAFLD³³. Histological samples were scored according to the NASH-Clinical Research Network (CRN) scoring system³⁴. In brief, the NAS was calculated as the sum of three histological components, including steatosis (grades 0–3), ballooning (grades 0–2) and lobular inflammation (grades 0–3). Patients with NAS of ≥ 5 correlated with Definite NASH, patients with NAS at 3 or 4 were correlated with borderline NASH, biopsies with scores of less than 3 were diagnosed as NAFL. The characteristics of the participants are listed in Supplementary Tables 3 and 4.

For cohort 3, serum and stool samples from 42 non-smokers diagnosed with biopsy-confirmed NAFLD, the diagnosis and scoring of NAFLD is described as above. The characteristics of the participants are listed in Supplementary Tables 5 and 6.

The exclusion criteria for cohort 1–3 were: diabetic ketoacidosis or hyperglycaemic hyperosmolar state; patients with decompensated cirrhosis; stage 3–5 chronic kidney disease; inflammatory bowel disease; cancer; pulmonary tuberculosis and AIDS; alcoholism; diseases judged by clinicians as unsuitable for biopsy; and taking antibiotics (including rifaximin), probiotics, prebiotics, proton-pump inhibitors and laxatives during the last 3 months. All clinical information was collected according to standard procedures. The collected metadata covered the participants' anthropometric features and information related to health status, age, sex and dietary habits. Peripheral blood samples were centrifuged at 1,000g and 4 °C for 5 min to obtain the serum. The levels of blood biochemical indicators (that is, triglycerides, TC, AST and ALT) were measured using an autoanalyzer (BioTek Instruments 800TS). Framingham risk score and 10-year-QRISK-3 score were calculated according previously reports^{35,36}. Faeces were collected in collection cups and immediately frozen at –80 °C until analysis. Mucosa biopsy samples were collected by colonoscopy from terminal ileum samples with no obvious abnormalities.

Animals

SPF mice of the C57BL/6J strain were randomly divided into different groups, housed 3–5 per cage and maintained under standard laboratory conditions with free access to rodent feed and water at the Department of Experimental Animal Science, Peking University Health Science Center or the National Cancer Institute, National Institutes of Health (NIH). Specifically, the light was on from 08:00 to 20:00, with the temperature kept at 21–24 °C and the humidity at 40–70%. GF C57BL/6J (GemPharmatech) were bred within sterile vinyl isolators and maintained at the gnotobiotic mouse facility at Department of Experimental Animal Science, Peking University Health Science Center. All of the animal experiments complied with the protocols for animal use, treatment and euthanasia approved by Peking University (permit: LA2020480) and the National Cancer Institute Animal Care and Use Committees of NIH (animal protocols: LM-016 and LM-096).

Prkaa1^{fl/fl} mice were purchased from the Jackson Laboratory (014141). To achieve intestinal-epithelium-specific AMPK α 1 disruption, *Prkaa1*^{fl/fl} mice

were crossed with Villin-cre mice (gifted by D. L. Gumucio) to obtain *Prkaa1*^{ΔIE} mice. Villin-cre mice express Cre recombinase in villus and crypt epithelial cells of the small and large intestines. The successful knockout of *Prkaa1* was assessed (Extended Data Fig. 4f). *Prkaa2*^{fl/fl} mice were purchased from GemPharmatech (T013159), and crossed with Villin-cre mice to obtain *Prkaa2*^{ΔIE} mice. The successful knockout of *Prkaa2* was assessed (Extended Data Fig. 4f).

Male mice (aged 8 weeks) were fed a HFHCD (Research Diets, D09100310). In assays of the drinking mouse model, SPF mice or GF mice were treated with nicotine (10 $\mu\text{g ml}^{-1}$, Sigma-Aldrich, 54-11-5) in sweet water containing 23.1 g l⁻¹ D-(–)-fructose (Sigma-Aldrich, F0127) and 18.6 g l⁻¹ D-(+)-glucose (Sigma-Aldrich, G8270). For assays of the mouse smoke exposure model, the mice were exposed to 60 mg m⁻³ cigarette smoke of total suspended particulate matter for 2 weeks and for 2 h per day. For the mouse subcutaneous injection model, the mice were administered nicotine (0.5 mg kg⁻¹) or saline daily for 2 weeks. For GW4869 (Selleck Chemicals, S7609) treatment, SPF mice were treated with control or 10 mg kg⁻¹ GW4869 by daily gavage. For ceramide (d18:1/16:0) (Avanti Polar Lipids, 860692) supplementation, littermate *Prkaa1*^{fl/fl} and *Prkaa1*^{ΔIE} mice were given vehicle or 10 mg kg⁻¹ ceramide (d18:1/16:0) by daily i.p. injection.

For bacterial colonization, SPF mice were treated with 2×10^8 CFU of *B. xylanisolvens* or *nicX*-knockout *B. xylanisolvens* in 200 μl of sterile anaerobic PBS by gavage every 3 days³⁷. WT or *nicX*-knockout *B. xylanisolvens* was grown anaerobically in BHI medium, and the cultures were washed and concentrated in anaerobic PBS to a final concentration of 1×10^9 CFU per ml under strict anaerobic conditions. The determination of CFU per ml was performed by a dilution-plate method using BHI medium containing 1% agarose under anaerobic conditions.

For analysis of mouse body-mass composition, the body fat content and fat ratio were measured and calculated using an MRI body-fat measurement instrument (EchoMRI-700, EchoMRI) 1 week before euthanasia. For biochemical analysis, hepatic and serum triglyceride and cholesterol and non-esterified fatty acid levels were determined using assay kits from Wako Diagnostics (Wako Chemicals) according to the manufacturer's instructions. ALT and AST levels were assessed using commercial ALT and AST assay kits (Catachem).

Quantitative MS

The concentrations of nicotine and HPB were measured by liquid chromatography coupled with tandem mass spectrometry (LC–MS/MS). For tissue analysis, tissue samples of approximately 0.1 g were homogenized after the addition of 10% methanol aqueous solution. Subsequently, 20 μl NaOH (1 M), 1 ml of ethyl acetate and 10 μl of the internal standard (500 ng ml⁻¹ nicotine-d4 for both nicotine and HPB) were added to 100 μl of the homogenates, and the mixtures were rapidly mixed and centrifuged at 13,000 rpm for 10 min. Subsequently, 900 μl of the supernatant was volatilized by nitrogen gas, and the sediment was dissolved in 100 μl of 70% aqueous acetonitrile (v/v). For serum and culture medium of the bacteria, 100 μl samples were prepared as described above. Then, 5 μl of the supernatant was analysed using the QTRAP 5500 system (AB SCIEX) with a Waters XBridge amide column (100 $\times$ 4.6 mm, inner diameter = 3.5 μm). The mobile phase was a mixture of acetonitrile (B) and water containing 0.1% formic acid and 2 mM ammonium formate (A) at a constant flow rate (0.5 ml min⁻¹). The proportion of mobile phase B was maintained at 25% by isocratic elution. The analytes [M+H]⁺ were measured at *m/z* 163.1–130.1 for nicotine, *m/z* 166.1–106.0 for HPB and *m/z* 167.1–134.1 for nicotine-d4. Lower relative s.d. values (less than 10%) were detected both between and within days. The recoveries of all the analytes were higher than 97%.

Metagenomic sequencing

Metagenomic sequencing was performed as previously described³⁸. DNA (700 ng per sample) was used for sample preparation. Sequencing libraries were generated using the Ultra DNA Library Prep Kit for

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Illumina; the manufacturer's recommendations and index codes were used to annotate sequences in each sample. The fragmented DNA ends were repaired, poly(A) tailed and ligated with a sequencing adaptor for Illumina sequencing. PCR amplification and DNA purification were performed using the AMPure XP system. The DNA concentration was measured using the Qubit DNA Assay Kit in the Qubit 2.0 fluorometer and diluted to 2 ng μl^{-1} . The library insert size was assessed using the Agilent Bioanalyzer 2100 system. Clustering of index-coded samples was performed on the cBot Cluster Generation System using the HiSeq 250 PE Cluster Kit according to the manufacturer's instructions. After cluster generation, the libraries were sequenced on the Illumina HiSeq 250 platform, and 150 bp paired-end reads were generated.

Metagenomic analysis

Sequencing data were converted to fastq files using bcl2fastq. Reads were then quality-control-trimmed using Trimmomatic³⁹ with the following parameters: PE -threads 10 -phred33 -validatePairs ILLUMINACLIP:TruSeq3-PE.fa:2:30:10 LEADING:3 TRAILING:3 MIN-LEN:50. We used MetaPhlAn2 (ref. ⁴⁰) for taxonomic analysis with the following parameters: --ignore_viruses --ignore_archaea --ignore_eukaryotes. The follow-up analysis was performed using MicrobiomeAnalyst⁴¹. The α -diversity was estimated on the basis of the gene profile of each sample according to the ACE, Chao1 and Shannon indices. The LEfSe analysis was performed using Galaxy⁴².

Microbial strains and gene-editing strategies

The *B. xylanisolvens* strain J1101 was isolated from human faeces as previously described⁴³. The strain was identified by comparing the 16S rRNA gene sequence with those in the NCBI reference database (<https://www.ncbi.nlm.nih.gov/>). The DNA fragments encoding full-length *nicX* (Supplementary Table 7) were cloned into the pET28a vector with a 6 $\times$ His tag at the N-terminal end using standard molecular cloning procedures. *nicX* was overexpressed in *E. coli* Rosseta (DE3), and the cells were grown at 37 °C to an OD₆₀₀ of 0.6. NicX protein was induced with 200 μM isopropyl- β -D-thiogalactoside (IPTG) overnight at 16 °C. An internal fragment (610 bp) of the *nicX* gene from *B. xylanisolvens* was cloned into the pGERM suicide vector containing *E. coli* (*bla*) and *B. xylanisolvens* (*ermG*) selective markers. The constructed vector was then transformed into the conjugative *E. coli* S17 strain. The donor *E. coli* S17 and recipient *B. xylanisolvens* were co-incubated under aerobic conditions, and the bacteria on the filter were transferred onto BHI medium agar plates supplemented (mutant selection) or not supplemented (control) with erythromycin (25 $\mu\text{g ml}^{-1}$) and gentamicin (200 $\mu\text{g ml}^{-1}$). Resistant colonies were then picked for identification by PCR using primers targeting junction regions between the pGERM and *nicX* genes. Details of the construction processes were reported previously⁴⁴. Both *B. xylanisolvens* and *nicX*-deficient *B. xylanisolvens* were grown in BHI medium supplemented with 0.3 g ml^{-1} cysteine under anaerobic conditions at 37 °C. *E. coli* and *nicX*-overexpressing *E. coli* were cultured in Luria broth (LB) medium at 37 °C for 24 h. To obtain bacteria in PBS for gavage, bacterial culture medium was centrifuged at 8,000g and 4 °C for 10 min and the pellets were then suspended in PBS (oxygen-free PBS for anaerobic bacteria).

Growth curves of *B. xylanisolvens*

B. xylanisolvens or *nicX*-deficient *B. xylanisolvens* (2×10^7 CFU per ml) were inoculated anaerobically into BHI medium at a ratio of 1:50 (v/v). To investigate the effect of nicotine on the growth of *B. xylanisolvens*, nicotine was added to the medium at the concentration of 100 $\mu\text{g ml}^{-1}$. Then, 200 μl of culture was taken every 2 h over a day to measure the optical density at 600 nm (OD₆₀₀) using the Spectra Max 190 microplate reader (Molecular Devices).

Microbial nicotine catabolism in vitro

B. xylanisolvens and *nicX*-deficient *B. xylanisolvens* were cultured in BHI medium containing 100 $\mu\text{g ml}^{-1}$ nicotine, and then incubated for 48 h at

37 °C anaerobically. The supernatants were sampled for MS detection of nicotine and HPB. WT *E. coli* and *nicX*-overexpressing *E. coli* were treated with 200 μM IPTG overnight at 16 °C to induce the expression of NicX protein before incubation. WT *E. coli* and *nicX*-overexpressing *E. coli* were cultured in LB medium and the other steps were the same as those described above.

Kinetic parameters for NicX were determined in assays containing 1 mM flavin mononucleotide, 25 mM Tris-HCl (pH 7.6) and nicotine. The reaction mixture was incubated for 20 min at 37 °C. Substrate nicotine concentrations varied from 6.16 μM to 789 μM , and a 20 μl sample was pipetted and mixed with an equal volume of acetonitrile to stop the reaction. Product formation was determined using LC-MS/MS, as described above. Reactions were performed in triplicate, and data were fitted to the Michaelis-Menten equation using the least-squares method.

Histological analysis

After euthanasia, liver tissues of mice were preserved in 10% formalin or embedded in OCT compound. Formalin-fixed paraffin-embedded liver tissue slides were stained with H&E and Sirius Red to assess lipid accumulation and liver-fibrosis, respectively. The frozen liver sections were stained with Oil Red O for visualizing lipid droplets. All of the processes were performed according to standard protocols followed by microscopy examination. The histology scores were evaluated according to the reported NAFLD scoring system³⁴.

Extraction, isolation and identification of HPB

The solvents used for extraction and chromatographic separation were of analytical grade. TLC was carried out on silica gel HSGF254, and the spots were visualized by spraying with Bismuth potassium iodide chromogenic agent. Silica gel and Sephadex LH-20 were used for column chromatography. HPLC separation was performed on an Agilent 1200 HPLC system using an ODS column (C_{18} , 250 $\times$ 9.4 mm; detector: UV) with a flow rate of 2.0 ml min^{-1} . NMR spectral data were obtained using the Bruker Avance-500 spectrometer.

The fermentation broth of *B. xylanisolvens* in BHI medium containing 100 $\mu\text{g ml}^{-1}$ nicotine (1 l) was extracted three times with ethyl acetate (3 $\times$ 500 ml), and the organic solvent was evaporated to dryness under a vacuum leaving the crude extract (565 mg). The crude extract was subsequently separated on a silica gel column (30.0 g of silica gel) eluted with hexane-ethyl acetate by gradient elution (v/v, 100:0, 100:2, 100:5, 100:10, 100:20, 100:40, each 500 ml) and then with dichloromethane-methanol (v/v, 100:1, 100:2, 100:5, 100:10, 0:100, each 500 ml) to give 10 fractions (BX-1-BX-10). Compound 1 (55.3 mg, R_f 18.3 min) was obtained from fraction BX-5 by reversed-phase high-performance LC using 23% methanol in water. The structure of the compound 1 was identified as the known compound 4-hydroxy-1-(3-pyridyl)-1-butanone (HPB) by comparing the experimental NMR data obtained with literature values⁴⁵.

Isolation and culture of intestinal primary cells

Intestinal primary cells were prepared from mice on the C57BL/6J background as previously described⁴⁶. Ileum (bottom third of the small intestine) was opened longitudinally, rinsed in cold PBS containing 1% penicillin-streptomycin and 2% glucose and the outer muscle layer was removed, then the tissue was chopped into approximately 5 mm pieces in cold PBS and further washed three times. After discarding the supernatant, 10 ml of digestion fluid (DMEM containing 0.1% type 1 neutral protease, 0.2% collagenase and 0.1% Y-27632) was added to the remaining ileum pieces, which were then incubated at 37 °C for 15 min with intermittent shocks. Stop buffer (DMEM containing 5% fetal bovine serum and 2% sorbitol) was added and the mixture filtered through 100 μm and 40 μm cell filters. Single cells were centrifuged at 500g at 4 °C for 5 min, and DMEM containing 10% FBS and 1% penicillin-streptomycin was used for ileal primary cell culture in an incubator at 37 °C under 5% CO_2 . Nicotine was applied to the ileal primary cells at a final concentration of 1 $\mu\text{g ml}^{-1}$.

Organoid isolation, culture and treatment

Isolation and culture of ileal organoids were performed as previously reported²⁶. In brief, ileum isolated from mice was dissected and washed ten times with Dulbecco's PBS. The ileal fragments were then incubated with Gentle Cell Dissociation Reagent (STEMCELL Technologies) to separate the crypts and villi from the intestinal basement membrane. After centrifugation, the crypts were isolated by filtration (70 µm) and resuspended in a 1:1 mixture of Matrigel (Corning) and IntestiCult organoid growth medium (STEMCELL Technologies) at a density of 6,000 crypts per ml. A droplet of 50 µl containing 300 crypts was placed into the centre of each well of a prewarmed 24-well plate, forming a dome. After the domes had solidified, 750 µl of organoid growth medium was added to each well. The crypts were cultured at 37 °C under 5% CO₂, and the medium was replaced every 3 days.

For compound C (AMPK inhibitor) treatment, ileal organoids were incubated with control, nicotine (1 µg ml⁻¹) or nicotine plus compound C (20 µM) for 12 h. For GW4869 treatment, HFHCD-fed WT mice were treated with nicotine water or nicotine water plus 10 mg kg⁻¹ GW4869 (by daily gavage) for 2 weeks, and organoids were then isolated and cultured. The ileal organoids were cultured for 7 days and treated with GW4869 (10 µM) and nicotine (1 µg ml⁻¹) for the last 3 days before being collected.

Lentiviral transduction and treatment in organoids

HFHCD-fed *Prkaa1*^{fl/fl} and *Prkaa1*^{ΔIE} mice were treated with nicotine water for 2 weeks, and ileal organoids were isolated. To further restore SMPD3 expression, isolated *Prkaa1*^{fl/fl} and *Prkaa1*^{ΔIE} organoids were infected with lentivirus (LV)-control or LV-Smpd3 (1 × 10⁶ PFU per 100 crypts), and the organoid-virus mixture was placed in an incubator at 37 °C for 1 h to allow transduction. After transduction, the infected organoids were plated and cultured for 7 days and treated with nicotine (1 µg ml⁻¹) for the last 3 days before being collected.

Targeted lipidomics analysis and ceramide quantification

Targeted lipidomics was performed according to a previous study with adjustments⁴⁷. Lipids included phosphatidylcholine, lyso-phosphatidylcholine, phosphatidylethanolamine, lyso-phosphatidylethanolamine, sphingomyelin and ceramide. Ileal tissue (20 mg) was added to 250 µl water and homogenized for 1 min, and the protein concentration determined by using the BCA assay. Then, 300 µl methanol and 1 ml methyl tert-butyl ether were added and vortexed for 10 s, centrifuged at 4 °C and 12,000 rpm for 10 min. The supernatant (95 µl) and internal standard solution (5 µl) was added into a 1.5 ml centrifuge tube and vortexed for 2 min. Subsequently, the mixture was transferred into a vial for MS detection. The Waters ACQUITY UPLC I-CLASS system with a Waters UPLC BEH C8 column (2.1 mm (inner diameter) × 100 mm (length), 1.7 µm (particle dimension)) was used for separation. For the positive-ion mode, the mobile phase consisted of acetonitrile:water (6:4, v/v; phase A) and isopropanol:acetonitrile (9:1, v/v, containing 5 mmol l⁻¹ ammonium formate and 0.1% formic acid; phase B) at a flow rate of 0.25 ml min⁻¹ and a column temperature of 55 °C, with an injection volume of 5.0 µl. The MS data under positive-ion mode were collected on the Waters XEVO TQ-S Micro system. The parameters were set as follows: ion source voltage, 3.0 kV; ion source temperature, 150 °C; desolvation temperature, 500 °C; desolvation gas flow, 1,000 l h⁻¹; cone voltage, 10 V; cone gas flow, 10 l h⁻¹. For data processing, the peak area was integrated by use of TargetLynx software with a retention time tolerance of 15 s. The final concentration was calculated by a single point internal standard method.

For the quantification of ceramides, 25 µl serum, 20 mg ileal tissue or organoids in 12-well plates were homogenized with chloroform and methanol (v/v, 2:1) containing 1 µM LM6002 (Avanti Polar Lipids) as the internal standard. The mixture was oscillated immediately and then centrifuged at 13,000 rpm for 20 min. The lower phase was dried

using a SpeedVac. The sediment was dissolved in 100 µl isopropanol and acetonitrile (v/v, 1:1) and analysed using the Eksigent LC100 system coupled with the AB SCIEX Triple TOF 5600 system using a Waters XBridge Peptide BEH C18 column (3.5 µm, 2.1 × 100 mm). The UPLC and MS parameters used were chosen according to a previous study²⁶. The lipid metabolites were quantified using MultiQuant 2.1 (AB SCIEX).

Phosphoproteomics

Phosphoproteomics analysis was performed as previously reported⁴⁸. In brief, ileal tissues were added to an appropriate amount of lysis buffer (including 4% SDS, 0.1 M Tris/HCl and 0.1 M dithiothreitol, pH 7.6) containing protease inhibitors and phosphatase inhibitors, and vortexed for 30 s. The samples were then ultrasonicated for 60 s (0.2 s on and 2 s off) with an amplitude of 25%. After standing at room temperature for 30 min, and centrifuging at 15,000 rpm for 10 min at 4 °C, the supernatant was carefully removed and centrifuged again. The protein concentration was measured using the BCA protein quantitative assay. Phosphopeptides were enriched using titanium dioxide (TiO₂). TiO₂ beads in tubes and TiO₂ microcolumns packed on centrifugation adaptors (GL Sciences, 5010–21514) were individually equilibrated with 100 µl of loading buffer (70% acetonitrile, 5% trifluoroacetic acid and 20% lactic acid (Sigma-Aldrich, L6661)). After centrifugation, the supernatants containing TiO₂ beads were removed. Peptides from all six subfractions in each experiment were diluted in 100 µl of loading buffer, mixed thoroughly with 5 mg of TiO₂ beads and incubated for 30 min. The mixtures of TiO₂ beads and peptides were then centrifuged at 800g for 5 min, the supernatants were transferred into new clear tubes for a second round of phosphopeptide enrichment, and the enriched TiO₂ beads were suspended in 100 µl of loading buffer, transferred into microcolumns and then packed into centrifugation adaptors. Each column with enriched TiO₂ beads was washed four times with 120 µl of loading buffer (150g, 10 min), washed once with 100 µl of washing buffer 1 (30% acetonitrile and 0.5% trifluoroacetic acid) (250g, 8 min) and washed twice with washing buffer 2 (0.4% trifluoroacetic acid and 80% acetonitrile) (250g, 5 min). Phosphopeptides were eluted with 150 µl of elution buffer (15% NH₄OH (Wako, 013–23355), 40% acetonitrile), and the first and second phosphopeptides enriched in each subfraction were combined, dried in a SpeedVac and stored at –80 °C. Lumos reverse-phase chromatography fusion was used for protein analysis. The volcano map was performed using the OmicStudio tools (<https://www.omicstudio.cn/tool>).

Cell culture and transient transfection

SW480 cells were grown in DMEM containing 100 U ml⁻¹ penicillin–streptomycin and 10% fetal bovine serum at 37 °C in a humidified incubator with 5% CO₂. Lipo3000 Transfection Reagent was used for the transfection of SW480 cells and the total DNA for each plate was adjusted using the relevant empty vector. Then, 24 h after transfection, the cells were collected. For all of the nicotine treatment cell experiments, nicotine was used at a concentration of 1 µg ml⁻¹. The cell lines were tested for mycoplasma contamination, and no cell lines were detected positive for mycoplasma contamination using a PCR-based method.

In vitro kinase assays

Bacterially purified SMPD3–His (86–655) proteins (2 µg) were incubated with partially purified rat liver active AMPK heterotrimer (α1–β1–γ2) (Promega, V4012) (0.5 µg) for 2 h at 30 °C in kinase buffer (25 mM HEPES, pH 7.4, 50 mM NaCl, 5 mM MgCl₂, 1 mM CaCl₂, 1 mM dithiothreitol, 30 µM ATP and 0.5 mg ml⁻¹ BSA) containing 1 µCi of [γ -³²P]ATP (Perkin Elmer). The reaction mixture was stopped by the addition of SDS sample buffer, and the labelled mixture was resolved by SDS–PAGE. The gels were dried, exposed to phosphor-imager screens for autoradiography, and stained with Coomassie Blue to visualize the proteins. Compound C was applied at a concentration of 20 µM.

Ubiquitination assay of SMPD3

The mutant proteins of SMPD3 (S209A, S209D, K103R or S209A/K103R) were constructed using a commercial point mutagenesis kit (Invitrogen, A14604) with seamless cloning and mutagenic primers. S209A: by a codon point mutation, the serine residue at position 209 of SMPD3 was changed to an alanine residue, which cannot be phosphorylated, thus mimicking SMPD3 with no phosphorylation at Ser209. S209D: by a codon point mutation, the serine residue at position 209 of SMPD3 was changed to an aspartate residue, which has a side chain similar to phosphorylated serine residue, thus mimicking SMPD3 with sustained phosphorylation at Ser209. K103R: by a codon point mutation, the lysine residue at position 103 of SMPD3 was changed to an arginine residue (R), which cannot be ubiquitinated, thus mimicking SMPD3 with no ubiquitination at Lys103.

For the SMPD3 ubiquitination assay, SMPD3–Flag (WT, K103R, S209A or S209A/K103R) was transfected into SW480 cells with Lipo3000 Transfection Reagent. Then, 24 h later, the cells were treated with DMSO or nicotine ($1 \mu\text{g ml}^{-1}$) for 24 h. The cells were washed with PBS and lysed in HEPES buffer (20 mM HEPES, pH 7.2, 50 mM NaCl, 1 mM NaF, 0.5% Triton X-100) plus 0.1% SDS and protease-inhibitor cocktail. The lysates were incubated with anti-Flag M2 affinity beads (Sigma-Aldrich, A2220) overnight at 4 °C. After extensive washing, proteins were eluted with 2× SDS sample buffer and resolved by SDS–PAGE and blotted with antibodies to Ub, Lys48–Ub or Lys63–Ub.

MS analysis

Modified SMPD3–Flag samples were prepared by immunoprecipitation from SW480 cells after DMSO or nicotine ($1 \mu\text{g ml}^{-1}$) treatment for 24 h. The SMPD3–Flag proteins were resolved by SDS–PAGE. The gel was stained with Coomassie Blue reagent and destained and the SMPD3–flag band was excised. The samples were subjected to reduction with 10 mM dithiothreitol for 30 min, alkylation with 55 mM iodoacetamide for 45 min, and in-gel digestion with trypsin. The digested samples were analysed by reversed-phase microcapillary–tandem mass spectrometry (LC–MS/MS) as described previously⁴⁹.

qPCR analysis

Liver tissues were flash-frozen in liquid nitrogen and stored at –80 °C. Total RNA from frozen liver tissues or ileal organoids was extracted using TRIzol reagent (Invitrogen). cDNA was synthesized from 2 μg of total RNA using 5× All-In-One RT MasterMix. A list of quantitative PCR (qPCR) primer sequences is provided in Supplementary Table 7. The relative amount of each mRNA was calculated after normalization to the corresponding *Actb* gene, and the results are expressed as fold changes relative to the control group.

Western blot analysis

Tissue and cell extracts were homogenized in ice-cold RIPA buffer with protease and phosphatase inhibitors and resolved on 8–10% SDS–PAGE gels and transferred to a polyvinylidene difluoride (PVDF) membrane. After blocking with 5% non-fat milk, the PVDF membrane was incubated with the desired primary antibody (diluted in TBST supplemented with 5% BSA) overnight at 4 °C on an orbital shaker with gentle shaking, followed by rinsing with TBST three times, 8 min each at room temperature, and then the secondary antibodies were incubated for 2 h at room temperature with gentle shaking. The secondary antibody (diluted in TBST supplemented with 5% BSA) was then removed, and the PVDF membrane was further washed three times with TBST for 8 min each at room temperature.

Antibodies

Rabbit polyclonal antibody to p-SMPD3(Ser209/208) (ABclonal, AP1312, 1:500 for IB) was raised and validated by Abclonal. Rabbit polyclonal antibody to ubiquitin–SMPD3(Lys103) (ABclonal, A21272, 1:500 for IB)

was raised and validated by Abclonal. Rabbit anti-p-AMPKα(Thr172) (Cell Signaling Technology, 2535, 1:1,000 for IB), anti-AMPKα (Cell Signaling Technology, 2532, 1:1,000 for IB), anti-AMPKα1 (Cell Signaling Technology, 2795, 1:1,000 for IB) anti-AMPKα2 (Cell Signaling Technology, 2757, 1:500 for IB), anti-ACTB (ABclonal, ab8227, 1:2,000 for IB), anti-SMPD3 (Santa Cruz Biotechnology, sc-166637, 1:500 for IB), anti-ubiquitin (Medical & Biological Laboratories, MK-11-3, 1:500 for IB), anti-ubiquitin (linkage-specific Lys48) (Cell Signaling Technology, 8081, 1:500 for IB), anti-ubiquitin (linkage-specific Lys63) (Cell Signaling Technology, 5621, 1:500 for IB) were used in this study.

Statistics

Heat maps and all charts were generated using GraphPad Prism software v.8.0. Metagenomic sequencing data were collected using the Illumina HiSeq PE250 system. Metabolite contents were collected by a QTRAP 5500 system. Phosphoproteome data were collected using the EASY-nano-LC1000 system. Other data were collected with Microsoft Excel 2019. All statistical data were analysed using SPSS v.26.0. The sample sizes were determined by power analysis using StatMate v.2.0. No data were excluded during the data analysis. All experimental data are reported as mean ± s.e.m. The Shapiro–Wilk normality test was used to determine the normal distribution of samples. Comparisons between two groups were performed using two-tailed unpaired Student's *t*-tests (normal distribution) or Mann–Whitney *U*-tests (non-normal distribution). Comparisons across multiple groups were assessed using one-way ANOVA (normal distribution) followed by Tukey's (with same s.d.) or Dunnett's T3 (with different s.d.) post hoc test, or Kruskal–Wallis test (non-normal distribution) followed by Dunn's test, as indicated in each figure legend.

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

All of the data supporting the findings of this study are included in the Article and Supplementary Tables 1–7. The metagenomic and phosphoproteomic data were uploaded to a public database (China National Microbiology Data Center (NMDC), under accession numbers NMDC10018157 and NMDC10018158, respectively). The following public databases were used in this study: NCBI reference database and Metaquery database. Source data are provided with this paper.

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Author contributions C.J. conceptualized and designed the study. B.C., L.S., G.Z., Z.S., K.W., L.Y., F.X., P.W., Y.D., Q.N., Q.W., Z.Z., J.X., J.L., Y. Luo., J.C., K.W.K., R.Z., Y.X. and M.-H.Z. performed the experiments and analysed the data. C.J., F.J.G., C.Y., Y. Li and M.-H.Z. supervised the study. B.C., L.S., G.Z. and C.J. wrote the manuscript with input from all of the authors. All of the authors edited the manuscript and approved the final manuscript.

Competing interests The authors declare no competing interests.

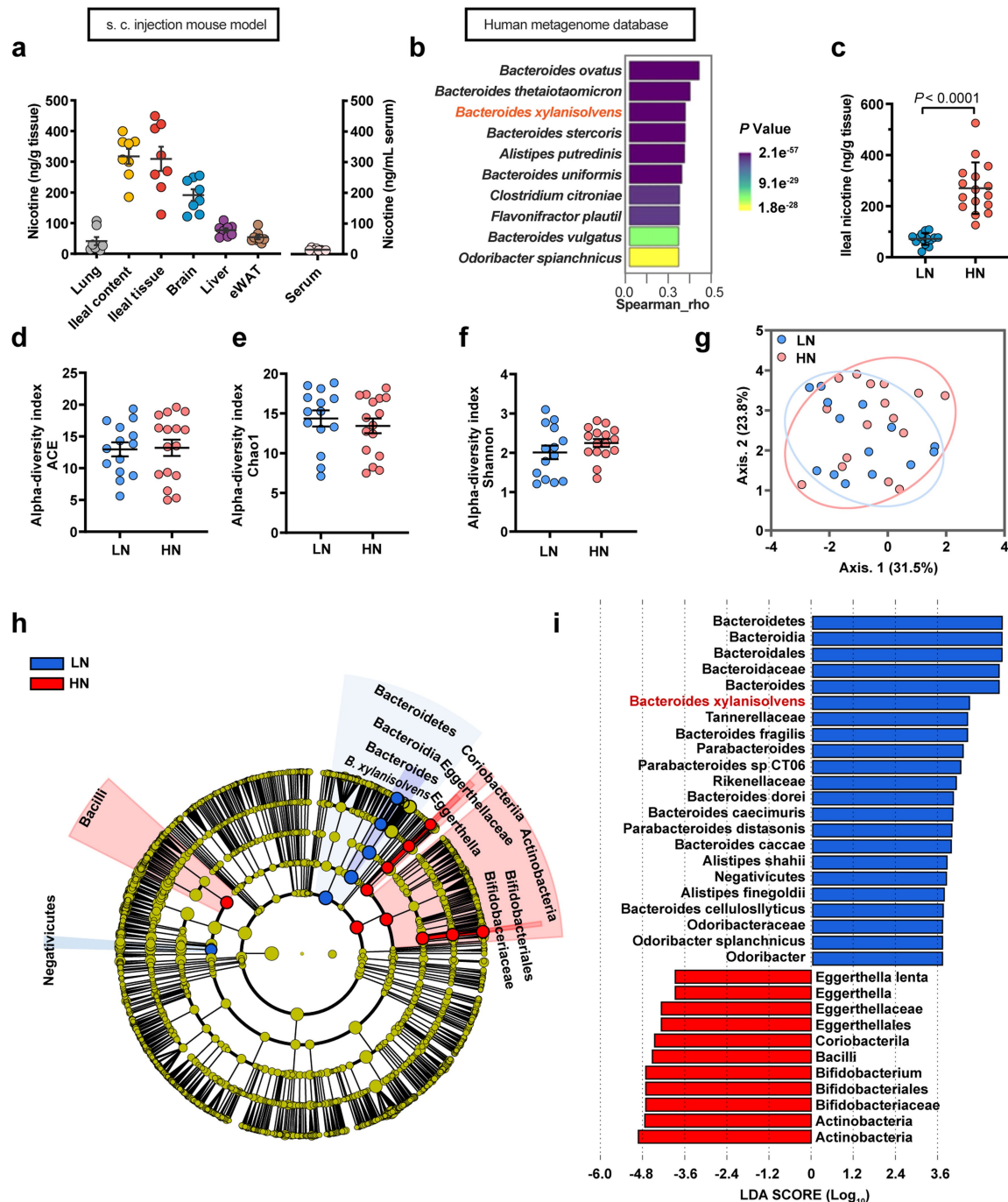
Additional information

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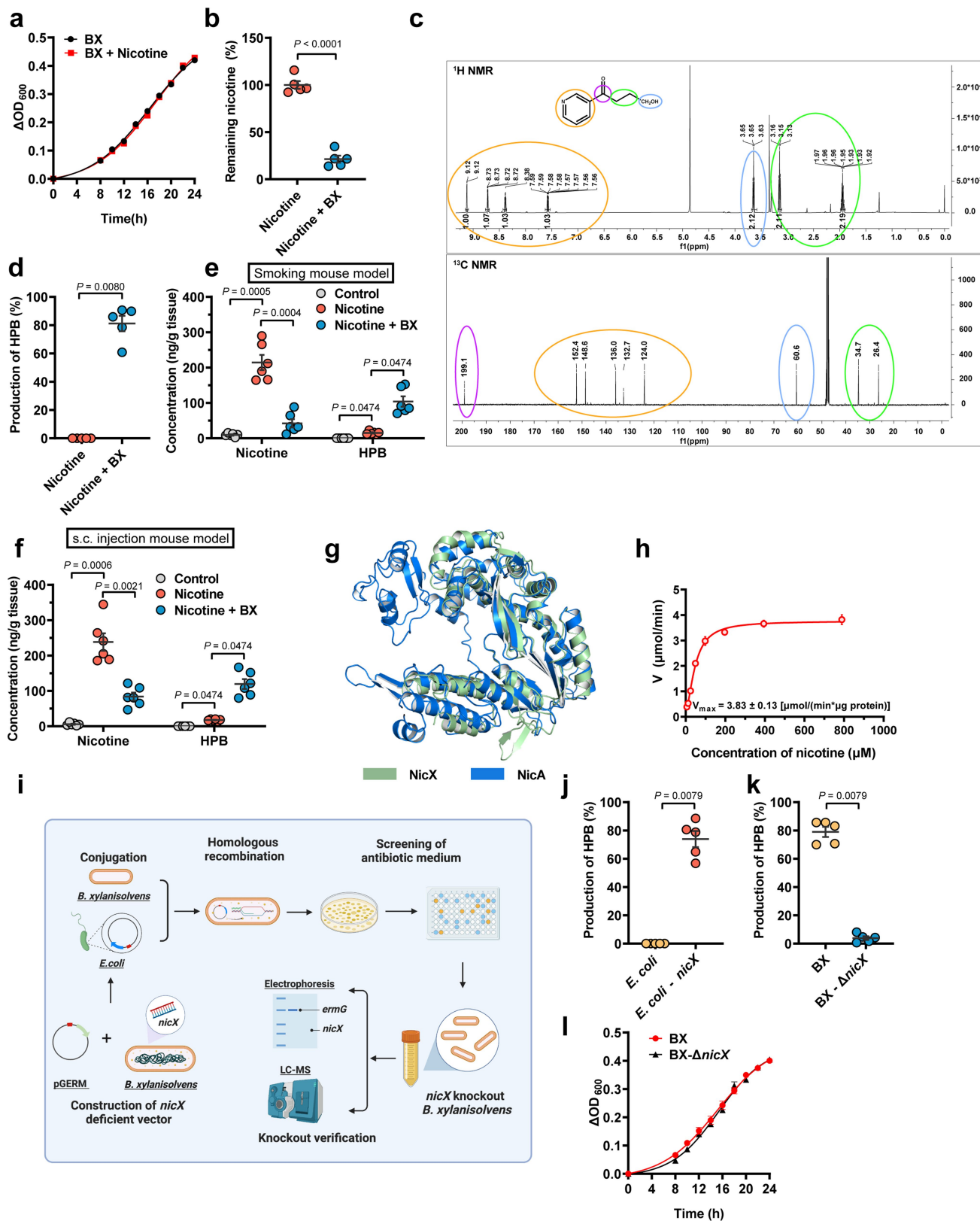
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Extended Data Fig. 1 | Gut microbiota composition differences between smokers with high nicotine and low nicotine levels. **a**, Quantification of the nicotine concentrations in lung, ileum content, ileum tissue, brain, liver, eWAT and serum samples obtained from s.c. injection mouse model for two weeks (SPF, $n = 8$ mice/group). **b**, Top 10 species of gut bacteria in humans that show high correlation with the known nicotine-degrading enzyme, *nicA*. These species were identified by MetaQuery. **c–i**, 30 smokers who were further divided into the HN (high nicotine, $n = 16$) and LN (low nicotine, $n = 14$) groups according to their ileal nicotine levels. Ileal nicotine concentrations in HN and LN groups. The data did not obey the normal distribution determined by the

Shapiro normality test; thus, the median was used as a break point, and divided into two groups (**c**). α -diversity of the gut microbiota between the LN and HN individuals, as indicated by the ACE (**d**), Chao1 (**e**) and Shannon indices (**f**). Partial least squares discriminant analysis (PLS-DA) using the Bray-Curtis distance (**g**). Taxonomic cladograms were generated by LefSe of metagenomic analysis data. The blue colour indicates enriched taxa in the LN group, and the red colour indicates enriched taxa in the HN group. The size of each circle is proportional to the taxon's abundance (**h**, **i**). Data are the means $\pm$ s.e.m. **b**, Correlations were assessed by nonparametric Spearman's test. **c–f**, Two-tailed Student's *t*-test.

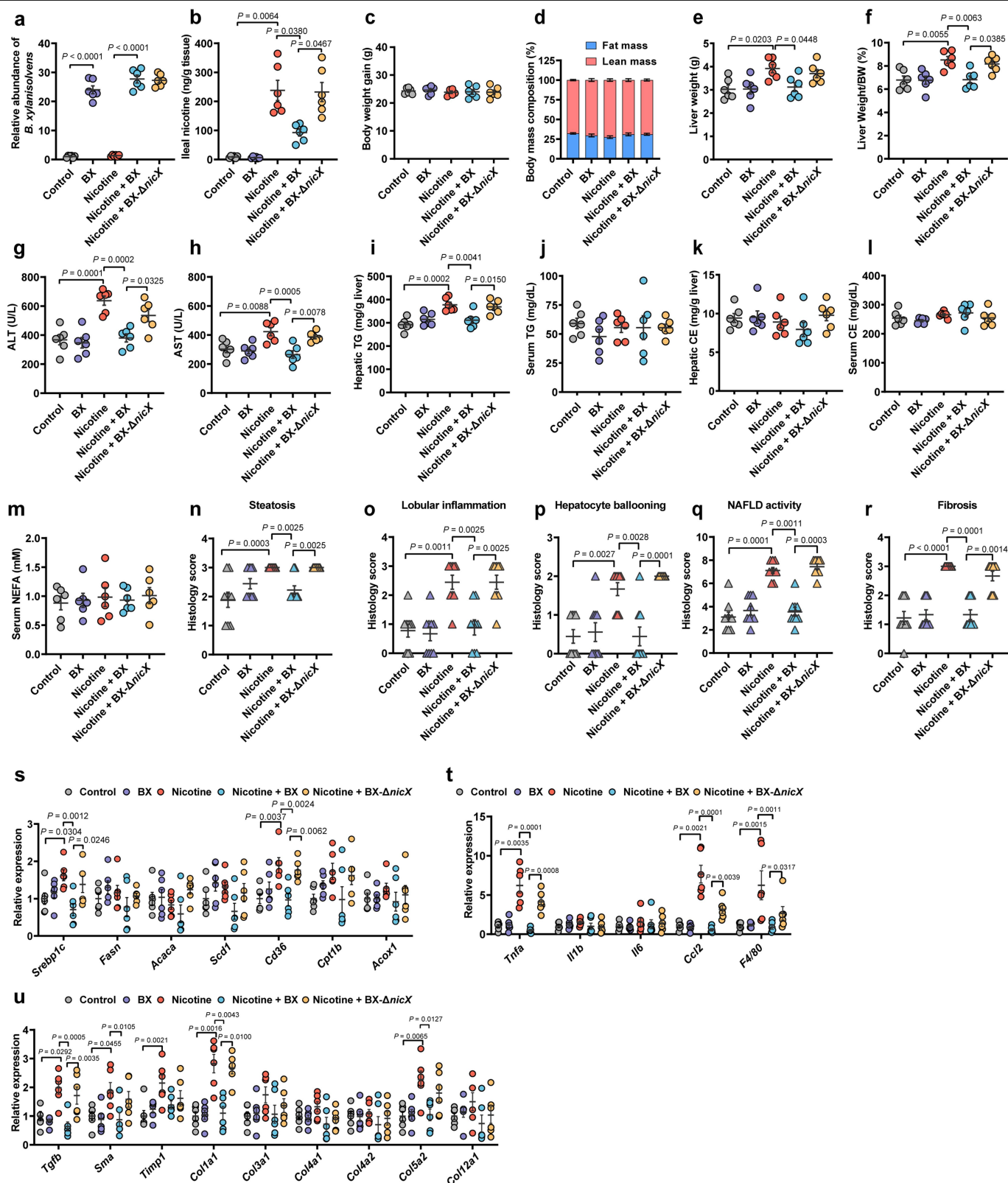


Extended Data Fig. 2 | See next page for caption.

Extended Data Fig. 2 | Identification of *B. xylanisolvens* as a nicotine

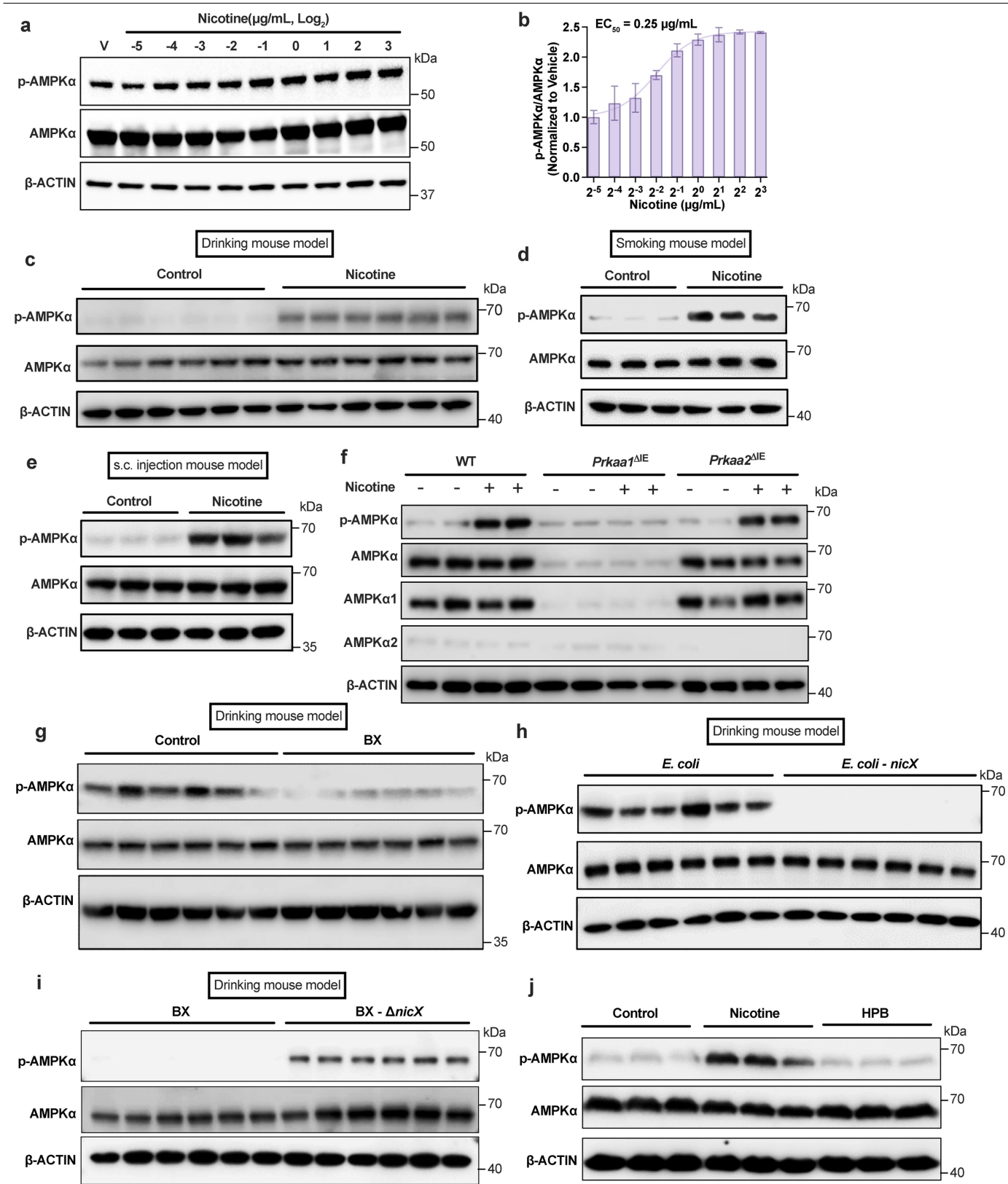
degrader. **a**, Growth curves of *B. xylanisolvens* with or without nicotine in culture medium (n = 3/group). **b**, Nicotine concentration in *B. xylanisolvens* in vitro cultivation compared with control (BHI medium with nicotine supplementation, n = 5/group). **c**, ¹H NMR spectrum (top) and ¹³C NMR spectrum (bottom) of HPB. **d**, Production of HPB in *B. xylanisolvens* in vitro supplementation compared with control (BHI medium with nicotine supplementation, n = 5/group). **e**, Nicotine and HPB concentrations in ileal tissues of the smoking exposure mouse model for two weeks (SPF, n = 6 mice/group). **f**, Nicotine and HPB concentration in ileal tissues of the subcutaneous injection mouse model for two weeks (SPF, n = 6 mice/group). **g**, Structural comparison of the SWISS-MODEL⁵⁰-predicted *B. xylanisolvens* NicX and predicted *Pseudomonas putida* NicA. The Root-Mean-Square-Deviation (RMSD) of 242 aligned residues is 1.323 Å. **h**, Nonlinear regression for nicotine

degradation catalysed by purified NicX. The reaction mixture contained 1 mM FMN, 25 mM Tris-HCl (pH 7.6), 20 ng NicX, and nicotine at different concentrations at 37 °C, n = 3/group. **i**, Schematic diagram illustrating the workflow for *nicX* gene deletion in *B. xylanisolvens*. The diagram was created using BioRender. **j**, Production of HPB in *E. coli* and *E. coli* + *nicX* in vitro cultivation (LB medium with nicotine supplementation, n = 5/group). **k**, Production of HPB in *B. xylanisolvens* and *B. xylanisolvens* Δ *nicX* in vitro cultivation in culture medium (BHI medium with nicotine supplementation, n = 5/group). **l**, Growth curves of WT and *nicX*-KO *B. xylanisolvens*. (n = 3/group). Data are the means $\pm$ s.e.m. **a**, **b**, **l**, Two-tailed Student's *t*-test. **d**, **j**, **k**, Two-tailed Mann-Whitney *U*-test. **e**, **f**, for Nicotine, one-way ANOVA with Dunnett's T3 post hoc test; for HPB, Kruskal-Wallis test with Dunn's test. Experiments in **a**, **b**, **d**, **h**, **j**, **k**, **l** were performed three times independently.



Extended Data Fig. 3 | *B. xylanisolvans* transplantation alleviates nicotine-accelerated NASH. HFHCD-fed SPF mice were treated with PBS, *B. xylanisolvans* colonization, nicotine water, nicotine water plus *B. xylanisolvans* colonization, and nicotine water plus *nicX* knock-out *B. xylanisolvans* colonization for 20 weeks ($n = 6$ mice/group). **a**, Faecal *B. xylanisolvans* abundance analyses of mice by qPCR. **b**, Ileal nicotine concentrations. **c**, Body weight gain. **d**, Body mass composition. **e**, Liver weights. **f**, Liver weight-to-body weight ratios. **g**, **h**, Serum ALT (**g**) and AST (**h**) levels. **i**, Hepatic TG content.

j, Serum TG content. **k**, Hepatic CE content. **l**, Serum CE content. **m**, Serum NEFA content. **n–r**, Histology scores of hepatic steatosis (**n**), lobular inflammation (**o**), ballooning (**p**), NAFLD activity (**q**), and fibrosis stage (**r**). **s–u**, Relative mRNA levels of genes related to hepatic lipid metabolism (**s**), inflammation (**t**) and fibrosis (**u**). Data are the means $\pm$ s.e.m. **c, e–i, k–m**, One-way ANOVA with Tukey's post hoc test. **a, b, j**, One-way ANOVA with Dunnett's T3 post hoc test. **d, n–u**, Kruskal-Wallis test with Dunn's test.

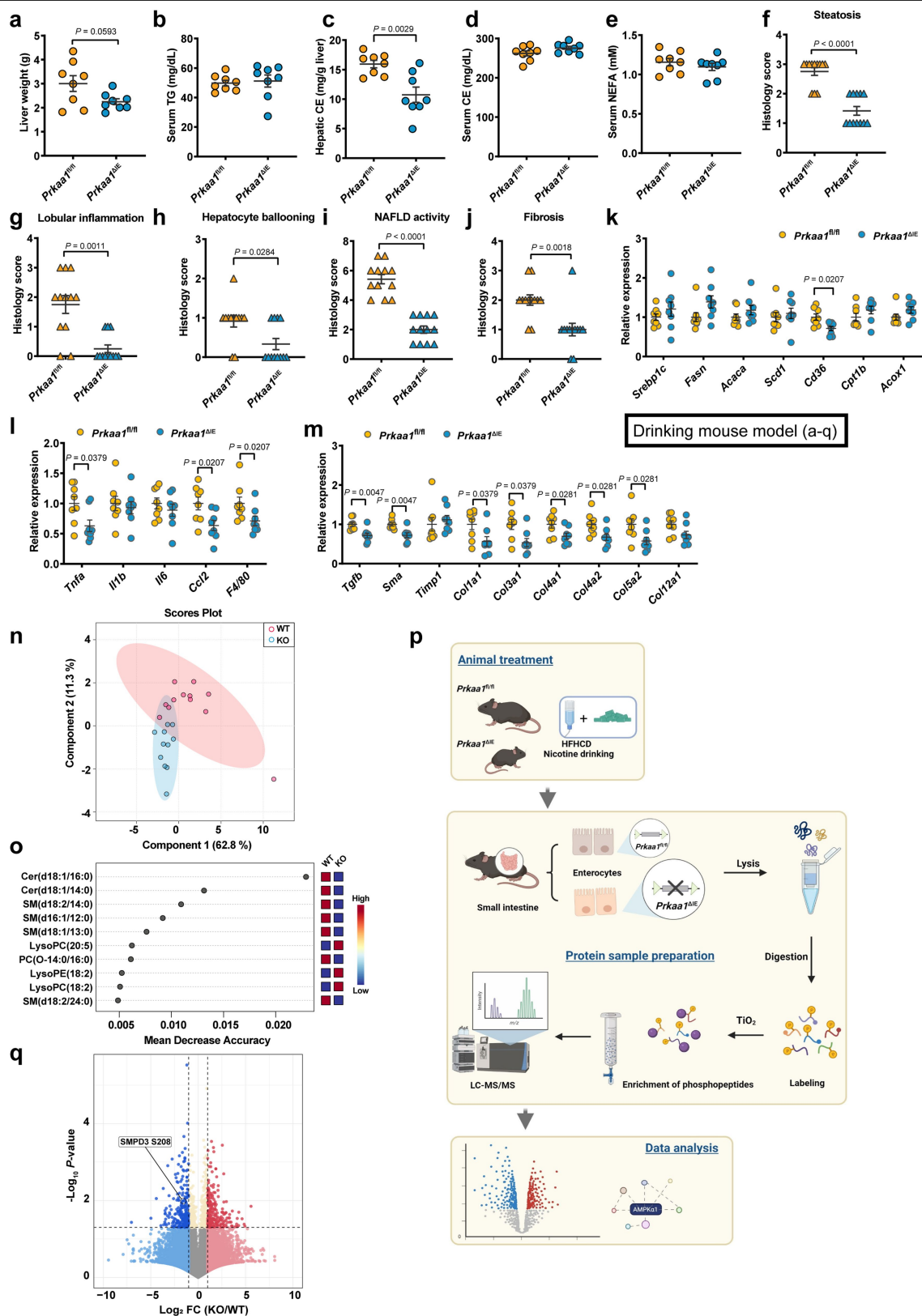


Extended Data Fig. 4 | See next page for caption.

Extended Data Fig. 4 | Nicotine-induced activation of intestinal AMPK α .

a, b, Activation of the AMPK α in ileal organoids after treatment with nicotine at different concentrations for 4 h (n = 3 independent experiments). **c**, Western blot analysis indicated that ileal AMPK α was activated in the nicotine drinking mouse model (SPF, n = 6 mice/group). **d, e**, Western blot analysis indicated that ileal AMPK α was activated in the smoking mouse model (**d**) and the subcutaneous injection mouse model (**e**) (SPF, n = 3 mice/group). **f**, Western blot analysis of ileal primary enterocytes isolated from WT, *Prkaa1*^{ΔIE} and *Prkaa2*^{ΔIE} mice (SPF) and then cultured with or without nicotine (1 $\mu\text{g ml}^{-1}$) treatment for 4 h. Experiments were performed with n = 4 mice/group.

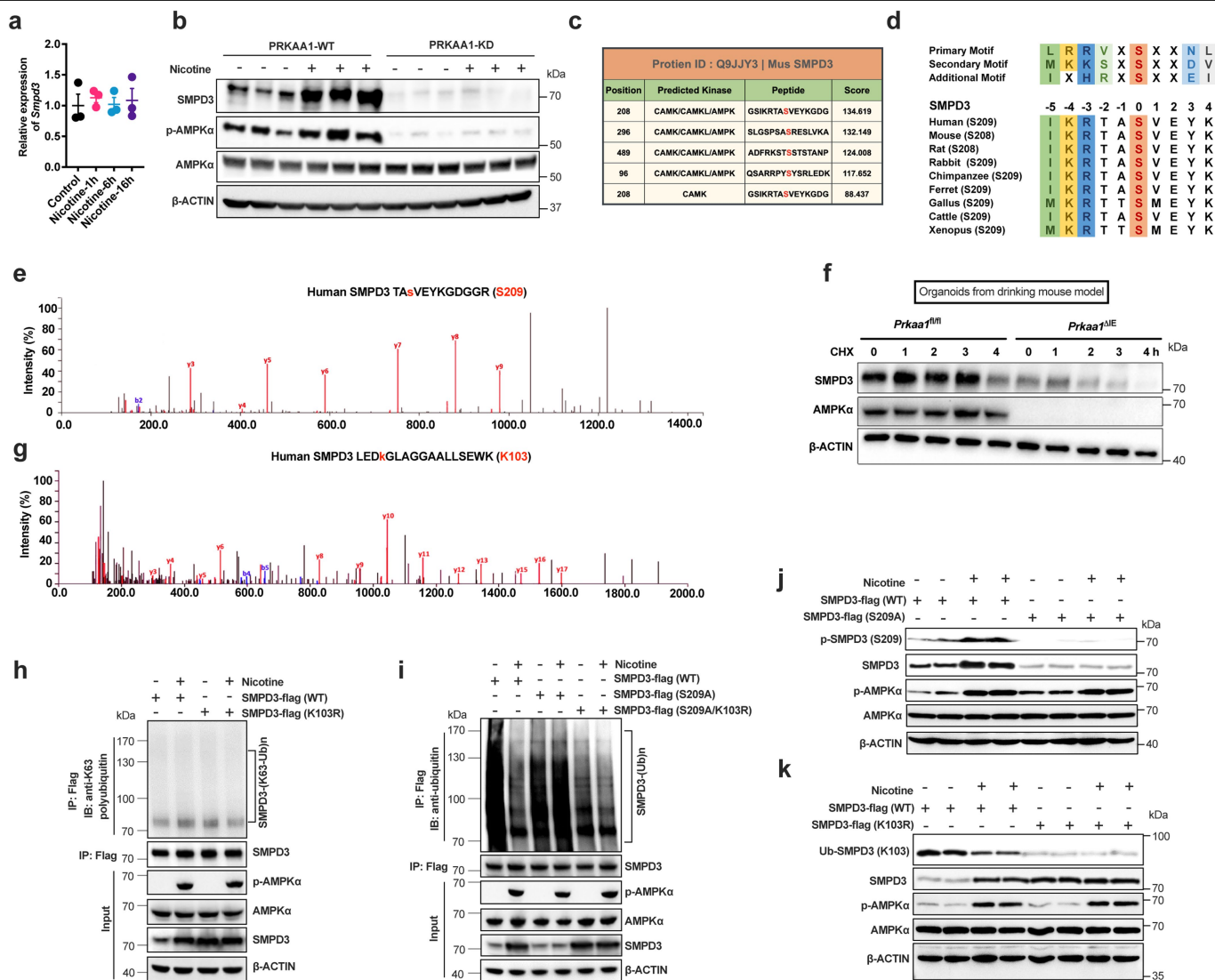
g, Western blot analysis showing ileal AMPK signalling in the nicotine drinking mouse model transplanted with Control or *B. xylanisolvens* (SPF, n = 6 mice/group). **h, i**, Western blot analysis showing ileal AMPK signalling in the nicotine drinking mouse model transplanted with *E. coli* or *nicX* knock-in *E. coli* (**h**); and WT or *nicX* knock-out *B. xylanisolvens* (**i**) (SPF, n = 6 mice/group). **j**, Western blot analysis showing AMPK signalling in SW480 cells incubated with nicotine (1 $\mu\text{g ml}^{-1}$) or HPB (1 $\mu\text{g ml}^{-1}$) for 4 h. This result is representative of 3 independent experiments. In **c–e**, **g–i**, mice were supplied nicotine plus HFHCD for two weeks. Data are the means $\pm$ s.e.m. (**b**).

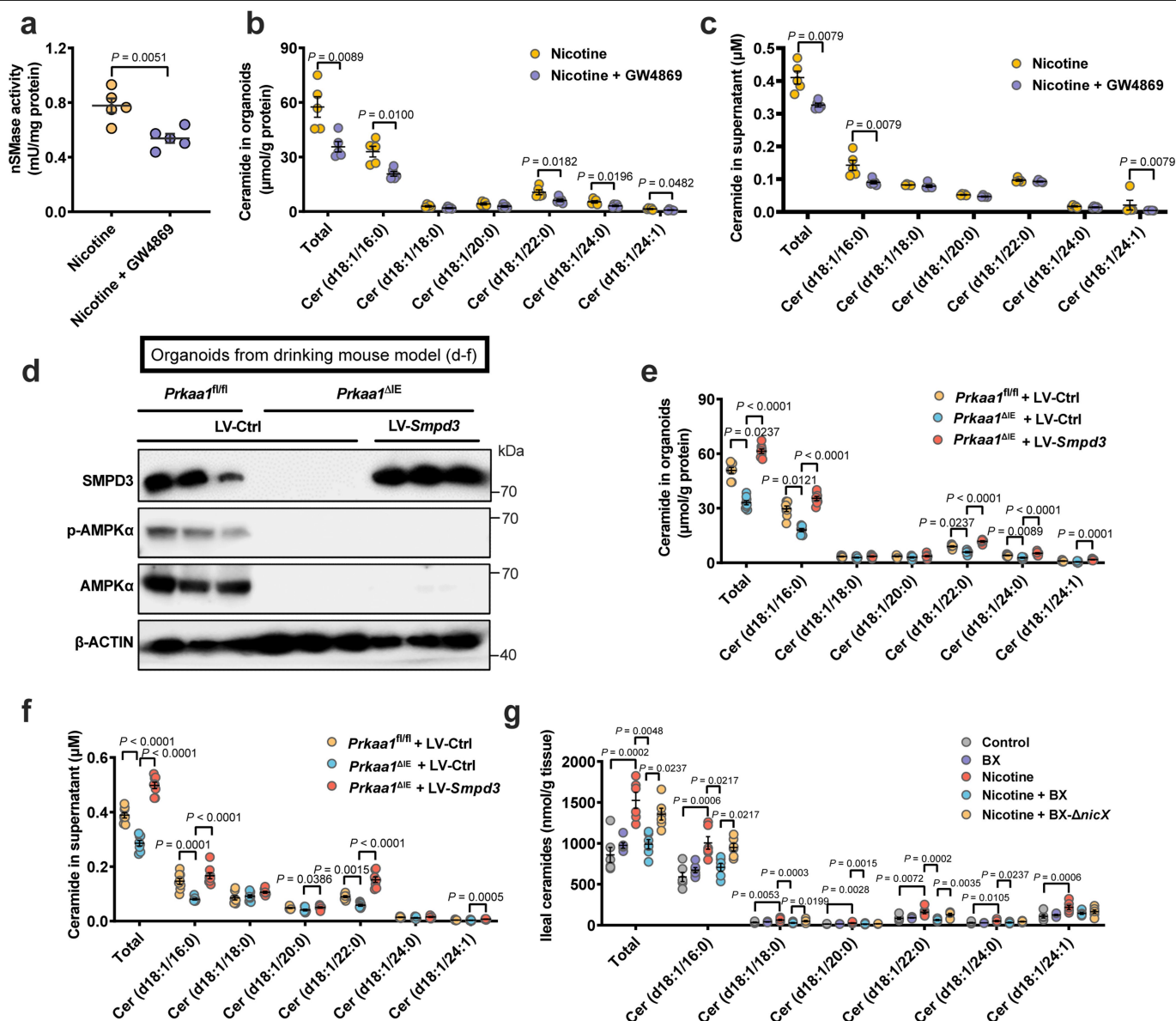


Extended Data Fig. 5 | See next page for caption.

Extended Data Fig. 5 | Intestinal AMPK α 1 deficiency improves NASH via decreasing ceramide generation. Eight-week-old male *Prkaa1^{fl/fl}* and *Prkaa1 ^{Δ IE}* mice were administered a HFHCD plus nicotine water for 20 weeks (SPF, n = 8 mice/group). **a**, Liver weight. **b–e**, Serum TG (**b**), hepatic CE (**c**), serum CE (**d**) and serum NEFA (**e**) contents. **f–j**, Histology scores of hepatic steatosis (**f**), lobular inflammation (**g**), ballooning (**h**), NAFLD activity (**i**), and fibrosis stage (**j**). **k–m**, Relative mRNA levels of genes related to hepatic lipid metabolism (**k**), inflammation (**l**) and fibrosis (**m**). **n, o**, Eight-week-old male *Prkaa1^{fl/fl}* (WT) and *Prkaa1 ^{Δ IE}* (KO) mice were administered a HFHCD plus nicotine water for 20 weeks (SPF, *Prkaa1^{fl/fl}*, n = 12 mice; *Prkaa1 ^{Δ IE}*, n = 11 mice).

PLS-DA analysis of lipid metabolites in the ileum (**n**). Random forest analysis showing the top 10 lipid metabolites that lead to differences in the ileal lipid profiles (**o**). **p**, A schematic diagram illustrating the workflow of phosphorylated proteomics. Created with BioRender. **q**, Volcano map of phosphorylated proteomics analysis from ileal epithelia of *Prkaa1^{fl/fl}* (WT) and *Prkaa1 ^{Δ IE}* (KO) mice (n = 5 mice/group) administered a HFHCD plus nicotine water for 20 weeks. Relative fold change (log2) of phosphorylated sites abundance was determined by comparing KO versus WT, and *P* values ($-\log_{10}$) were calculated by two-tailed *t*-test. Data are the means $\pm$ s.e.m. **a–e**, Two-tailed Student's *t*-test. **f–m**, Two-tailed Mann-Whitney *U*-test.



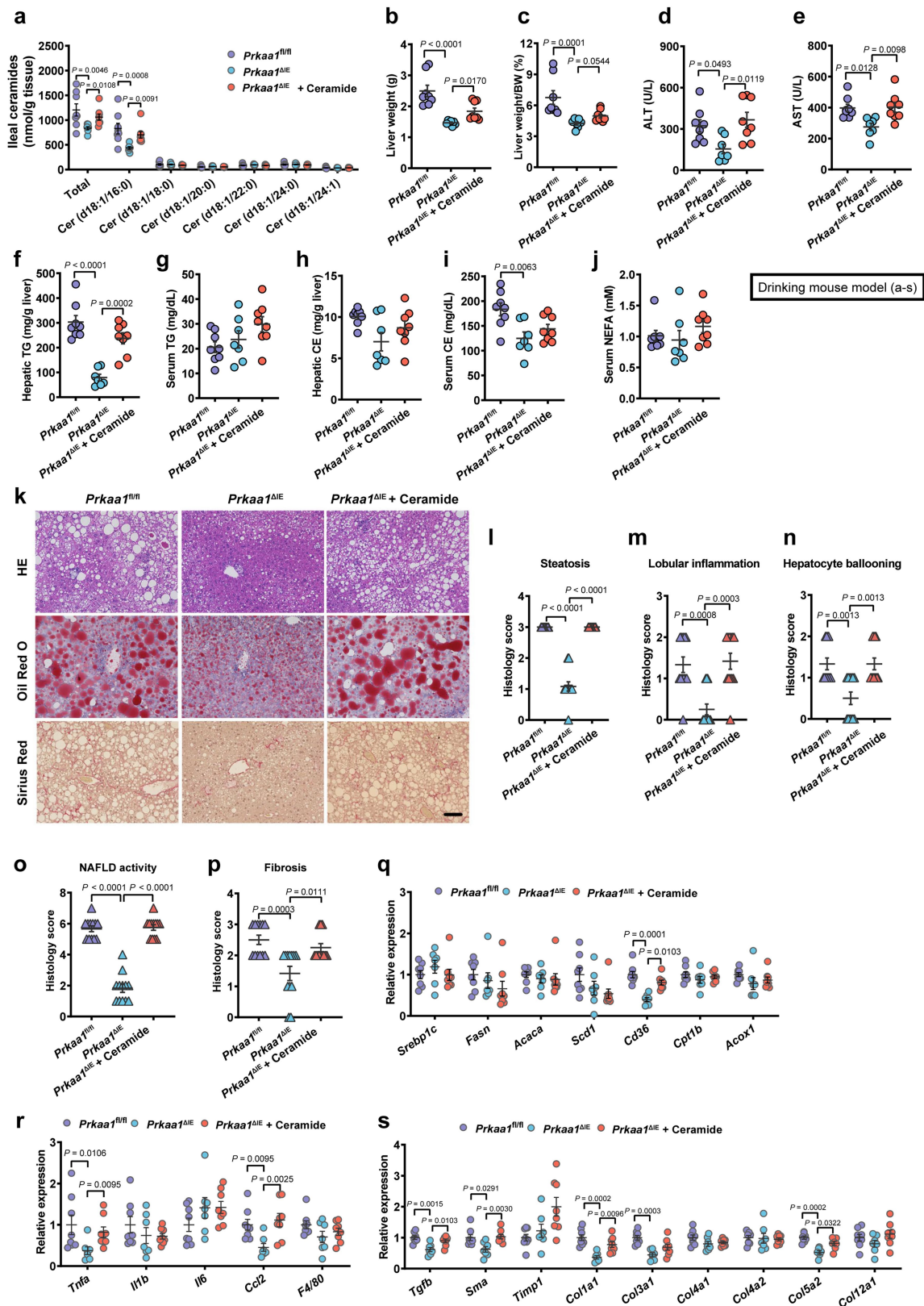


Extended Data Fig. 7 | Interaction between p-AMPKα and SMPD3 in intestinal ceramide production.

a-c, HFHCD-fed SPF mice were treated with nicotine water or nicotine water plus 10 mg/kg GW4869 (by daily gavage) for 2 weeks, and ileal organoids were then isolated and cultured for 7 days and treated with GW4869 (10 μM) and nicotine (1 μg ml⁻¹) for the last 3 days before the detection of ceramide production and secretion (n = 5 mice/group).

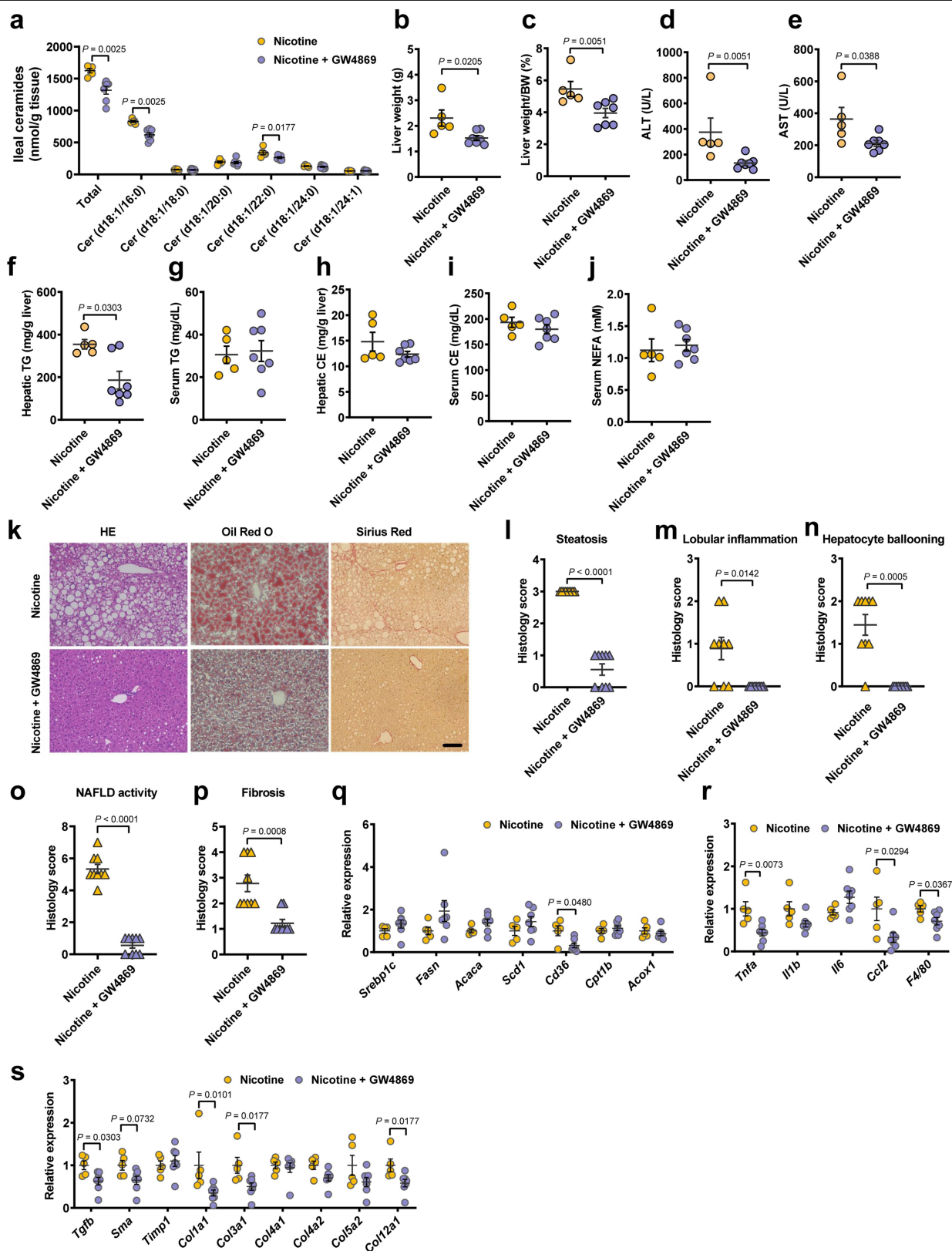
a, nSMase activity. **b**, Ceramide profiles in isolated organoids. **c**, Ceramide profiles in the supernatant of isolated organoids. **d-f**, HFHCD-fed *Prkaa1^{fl/fl}* and *Prkaa1^{ΔIE}* mice (SPF) were treated with nicotine water for 2 weeks, and organoids were then isolated and infected with LV (lentivirus)-Ctrl or LV-Smpd3, the infected organoids were plated and cultured for 7 days and treated with nicotine (1 μg ml⁻¹) for the last 3 days before the detection of

ceramide production and secretion. Western blot analysis for verifying SMPD3 overexpression. (n = 3 mice/group) (**d**). Ceramide profiles in isolated organoids. (n = 8 mice/group) (**e**). Ceramide profiles in the supernatant of isolated organoids. (n = 8 mice/group) (**f**). **g**, HFHCD-fed WT mice were transplanted with PBS, *B. xylanisolvens* colonization, nicotine water, nicotine water plus *B. xylanisolvens* colonization, and nicotine water plus *nicX* knock-out *B. xylanisolvens* colonization for 20 weeks (SPF, n = 6 mice/group), and ileal tissues were collected for ceramide profile analysis. Data are the means ± s.e.m. **a, b**, Two-tailed Student's *t*-test. **c**, Two-tailed Mann-Whitney *U*-test. **f**, One-way ANOVA with Tukey's post hoc test. **e, g**, Kruskal-Wallis test with Dunn's test.



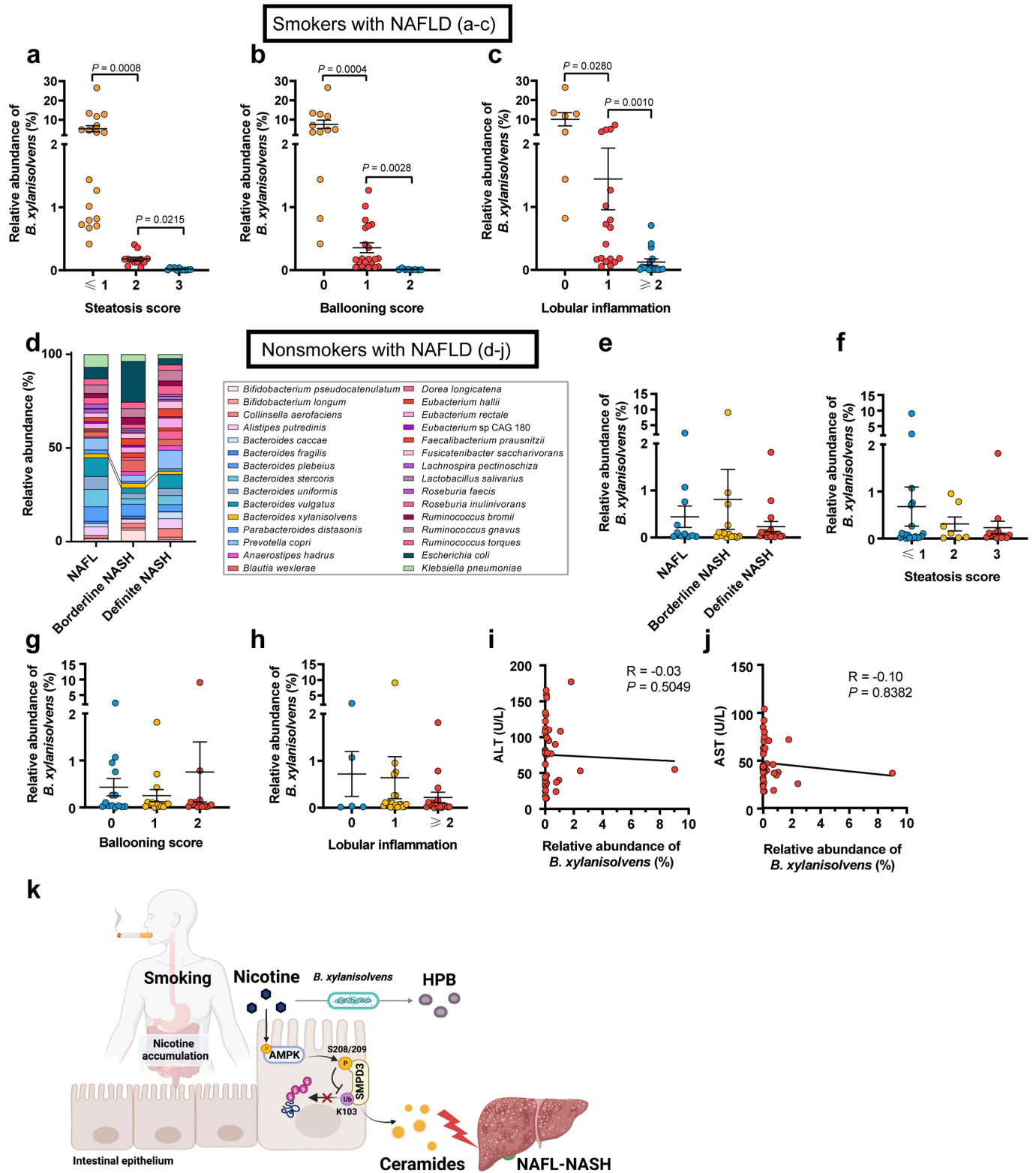
Extended Data Fig. 8 | Ceramide supplementation eliminates the beneficial effects derived from intestinal AMPK α 1 deficiency. Eight-week-old male *Prkaa1^{fl/fl}* and *Prkaa1^{ΔIE}* mice were treated with or without 10 mg/kg ceramide (d18:1/16:0) by daily i.p. injection under HFHCD plus nicotine water treatment for 20 weeks (SPF, *Prkaa1^{fl/fl}*, $n = 8$ mice; *Prkaa1^{ΔIE}*, $n = 7$ mice; *Prkaa1^{ΔIE}* + Ceramide, $n = 8$ mice). **a**, Ileal ceramide profiles. **b**, Liver weights. **c**, Liver weight-to-body weight ratios. **d**, **e**, Serum ALT (**d**) and AST (**e**) levels. **f**–**j**, Hepatic TG (**f**), serum TG (**g**), hepatic CE (**h**), serum CE (**i**) and serum NEFA

(**j**) contents. **k**, Representative H&E staining (upper), Oil Red O staining (middle) and Sirius Red staining (lower) of liver sections ($n = 4$ mice/group, 3 images/mouse). Scale bar, 100 μ m. **l**–**p**, Histology scores of steatosis (**l**), lobular inflammation (**m**), hepatocyte ballooning (**n**), NAFLD activity (**o**) and fibrosis stage (**p**). **q**–**s**, Relative mRNA levels of genes related to hepatic lipid metabolism (**q**), inflammation (**r**) and fibrosis (**s**). Data are the means $\pm$ s.e.m. **d**–**g**, **i**, One-way ANOVA with Tukey's post hoc test. **h**, One-way ANOVA with Dunnett's T3 post hoc test. **a**–**c**, **j**, **l**–**s**, Kruskal-Wallis test with Dunn's test.



Extended Data Fig. 9 | Inhibition of SMPD3 ameliorates nicotine-induced NASH. Eight-week-old male SPF mice were randomly grouped and administered vehicle or 10 mg/kg GW4869 by daily gavage under HFHCD plus nicotine water treatment for 20 weeks (Nicotine, $n = 5$ mice; Nicotine + GW4869, $n = 7$ mice). **a**, Ileal ceramide profiles. **b**, Liver weights. **c**, Liver weight-to-body weight ratios. **d**, **e**, Serum ALT (**d**) and AST (**e**) levels. **f**, **j**, Hepatic TG (**f**), serum TG (**g**), hepatic CE (**h**), serum CE (**i**) and serum NEFA (**j**) contents. **k**, Representative

H&E staining (left), Oil Red O staining (middle) and Sirius Red staining (right) of liver sections ($n = 3$ mice/group, 3 images/mouse). Scale bar, 100 μ m. **l**–**p**, Histology scores of steatosis (**l**), lobular inflammation (**m**), hepatocyte ballooning (**n**), NAFLD activity (**o**) and fibrosis stage (**p**). **q**–**s**, Relative mRNA levels of genes related to hepatic lipid metabolism (**q**), inflammation (**r**) and fibrosis (**s**). Data are the means $\pm$ s.e.m. **b**, **e**, **g**–**j**, **r**, Two-tailed Student's t -test. **a**, **c**, **d**, **f**, **l**–**q**, **s**, Two-tailed Mann-Whitney U -test.



Extended Data Fig. 10 | *B. xylanisolvans*-mediated nicotine degradation negatively correlates with clinical NASH. In 41 smokers with NAFLD, NAFL $n = 11$, borderline NASH $n = 16$, definite NASH $n = 14$. **a–c**, Relative abundances of *B. xylanisolvans* associated with steatosis score (**a**), ballooning score (**b**), and lobular inflammation (**c**) in smokers with NAFLD. In 42 non-smokers with NAFLD, including NAFL ($n = 11$), borderline NASH ($n = 14$), and definite NASH ($n = 17$). **d**, Bacterial taxonomic profiling of the gut microbiota from non-smokers with different NAFLD stages at the species level. **e**, Relative abundances of *B. xylanisolvans* associated with different NAFLD stages in

non-smokers with NAFLD. **f–h**, Relative abundances of *B. xylanisolvans* associated with steatosis score (**f**), ballooning score (**g**), and lobular inflammation (**h**) in non-smokers with NAFLD. **i, j**, Correlative analysis of *B. xylanisolvans* with ALT (**i**) and AST (**j**). Correlations between variables were assessed by linear regression analysis. Linear correlation index R square and P values were calculated. **k**, Summary diagram illustrating the role of microbial nicotine degradation in ceramide modulation and NAFL-NASH progression. Created with BioRender. Data are the means $\pm$ s.e.m. **a–c, e–h**, Kruskal-Wallis test with Dunn's test.

Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
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- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Metagenomic sequencing data were collected by Illumina HiSeq PE250. Metabolite content were collected by a QTRAP 5500 system. Phosphoproteome data were collected by EASY-nano-LC1000. TopSpin (version 3.2) was used to collect NMR data. Other data were collected with Microsoft Excel 2019.
Data analysis	Heatmap and all charts were generated using GraphPad Prism software 8.0. SPSS version 26.0 was used for statistical analysis. The sample sizes were determined by power analysis using StatMate version 2.0. Lumos reversed- phase chromatography fusion was used for protein analysis. The volcano map was performed using the OmicStudio tools. The lipid metabolites were quantified using MultiQuant 2.1 software and TargetLynx software. We used MetaPhlAn2 for metagenomic taxonomic analysis.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

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All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All the data that support the findings of this study are included in the published article and Supplementary Tables. The metagenomic and phosphoproteomic data were uploaded to public database (China National Microbiology Data Center (NMDC), with accession numbers as NMDC10018157, and NMDC10018158, respectively). The following public databases were applied in this study: NCBI reference database, Metaquery database.

Human research participants

Policy information about [studies involving human research participants and Sex and Gender in Research](#).

Reporting on sex and gender

The current study mainly focused on the effect of tobacco use on NALFD/NASH. Considering the high proportion of male smokers (PMID: 31737583, the ratio of male and female smokers is 95:3), we included 143 male (by sex) participants for the analysis. The information regarding the sex of each participant was collected by self-reporting via a questionnaire.

Population characteristics

Cohort1: terminal ileal mucosal biopsy, serum and stool samples were collected from 30 unrelated non-smokers and 30 unrelated smokers. All the subjects were diagnosed with colon polyps. During the polypectomy under a colonoscopy, we obtained a trace amounts of terminal ileal mucosa tissue with no obvious abnormalities.
Cohort 2: serum and stool samples from 41 unrelated smokers diagnosed with biopsy-confirmed NAFLD.
Cohort 3: serum and stool samples from 42 unrelated nonsmokers diagnosed with biopsy-confirmed NAFLD.
The exclusion criteria for cohort 1, 2 and 3 were: diabetic ketoacidosis or hyperglycemic hyperosmolar state; patients with decompensated cirrhosis; stage 3-5 chronic kidney disease; inflammatory bowel disease; cancer; pulmonary tuberculosis and AIDS; alcoholism; diseases judged by clinicians as unsuitable for biopsy; taking antibiotics (including rifaximin), probiotics, prebiotics, proton pump inhibitors, and laxatives during the last 3 months.

Recruitment

For cohort 1, 30 unrelated non-smokers and 30 unrelated smokers healthy individuals were recruited.
For cohort 2, 41 unrelated smokers diagnosed with biopsy-confirmed NAFLD were recruited.
For cohort 3, 42 unrelated nonsmokers diagnosed with biopsy-confirmed NAFLD were recruited.
All of the participants provided written informed consent.
The exclusion criteria for cohort 1-3 were: diabetic ketoacidosis or hyperglycemic hyperosmolar state; patients with decompensated cirrhosis; stage 3-5 chronic kidney disease; inflammatory bowel disease; cancer; pulmonary tuberculosis and AIDS; alcoholism; diseases judged by clinicians as unsuitable for biopsy; taking antibiotics (including rifaximin), probiotics, prebiotics, proton pump inhibitors, and laxatives during the last 3 months.

Ethics oversight

The study was approved by the Ethical Committee of the First Affiliated Hospital, Zhejiang University, School of Medicine for the cohort 1 [2021(875)] and by the Ethical Committee of the First Affiliated Hospital of Wenzhou Medical University for the cohort 2 & 3 [2016-246]. In addition, informed written consent was obtained from all subjects.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

There were at least 3 samples or replicates for each experiment in this study. For animal experiments, there were over 5 mice per group in order to obtain statistical significance, which were similar to those previously used by us and others in this field (ref. 9, 21, 26, 37).

Data exclusions

No data was excluded.

Replication

All experimental findings were repeated at least three times as stated in figure legends.

Randomization

Human subjects were randomly selected, and divided into different groups according to self-reported smoking status. 8 week old mice were divided at random into experimental groups, with at least 5 mice per group, and the groups did not present differences

in body weights before the treatments.

For sample testing (mass spectrometry analysis, biochemical analysis (such as TG and CE)), samples were processed tested in random orders. In cell experiments, cells of each group were parallel seeded and randomly assigned to different treatments.

Blinding

Investigators were blinded to group allocation during data collection and analysis. During the treatment of live animals, cages during sample collection and processing were labeled as code names that were later revealed by the individuals who picked and treated animals, but did not participate in sample collection and processing.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Anti-phospho-AMPKa-T172 (Cell Signaling Technology, cat. #2535,1:1000 for IB),
 Anti-AMPKa (Cell Signaling Technology, cat. #2532,1:1000 for IB),
 Anti-AMPKal (Cell Signaling Technology, cat. #2795,1:1000 for IB)
 Anti-AMPKa2 (Cell Signaling Technology, cat. #2757,1:500 for IB),
 Anti-ACTB (ABclonal, cat. #ab8227,1:2000 for IB),
 Anti-SMPD3 (Santa Cruz Biotechnology, cat. # sc-166637,1:500 for IB),
 Anti-Ubiquitin (Medical & Biological Laboratories Co., Ltd, cat. #MK-11-3, 1:500 for IB),
 Anti-Ubiquitin (linkage-specific K48) (Cell Signaling Technology, cat. #8081, 1:500 for IB),
 Anti-Ubiquitin (linkage-specific K63) (Cell Signaling Technology, cat. #5621, 1:500 for IB),
 Anti-phospho-SMPD3-Ser209 (ABclonal, cat. #AP1312, 1:500 for IB),
 Anti-ubiquitin-SMPD3-K103 (ABclonal, cat. #A21272, 1:500 for IB).

Validation

Anti-phospho-AMPKa-T172 (cat. #2535), citation: J Biol Chem. 2021;296:100075. doi:10.1074/jbc.RA120.016193
 Anti-AMPKa (cat. #2532), citation: J Biol Chem. 2007;282(45):32549-32560. doi:10.1074/jbc.M706536200
 Anti-AMPKal (cat. #2795), citation: Nat Med. 2015;21(4):373-382. doi:10.1038/nm.3826
 Anti-AMPKa2 (cat. #2757), citation: Nat Med. 2015;21(4):373-382. doi:10.1038/nm.3826
 Anti-ACTB (cat. #ab8227), citation: J Clin Invest. 2020;130(l):438-450. doi:10.1172/JCI130976
 Anti-SMPD3 (cat. # sc-166637), citation: J Clin Invest. 2021;131(9):el42865. doi:10.1172/JCI142865
 Anti-Ubiquitin (cat. #MK-11-3), citation: J Biol Chem. 2004 ;279(51):53533-43. doi: 10.1074/jbc.M402916200
 Anti-Ubiquitin (linkage-specific K48) (cat. #8081), citation: Immunity. 2021 ;54(2):247-258.e7. doi: 10.1016/j.immuni.2020.11.020
 Anti-Ubiquitin (linkage-specific K63) (cat. #5621), citation: Nat Cell Biol. 2021 ;23(9):978-991. doi: 10.1038/s41556-021-00732-8
 Anti-phospho-SMPD3-Ser209 antibody (cat. #AP1312, raised and validated by the manufacturer: <https://abclonal.com.cn/catalog/AP1312>) purchased from Abclonal was validated in this study using SMPD3-S209A mutation protein transfected in cell lines.
 Anti-ubiquitin-SMPD3-K103 antibody (cat. #A21272, raised and validated by the manufacturer: <https://abclonal.com.cn/catalog/A21272>) purchased from Abclonal was validated in this study using SMPD3-K103R mutation protein transfected in cell lines.

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)

SW480 cell: CCL-228 from ATCC (Manassas, VA; Lot Number: 70043599).

Authentication

SW480 cell is newly purchased from ATCC, and all ATCC cell lines undergo authentication tests during the accessioning process. This process is described in the online ATCC brochure "Maintaining High Standards in Cell Culture". Observations of recovery and growth are recorded along with morphological appearance. Although no detailed authentication information by authors, the strain was used at very low passages in the study and no phenotype was analyzed from the cell line.

Mycoplasma contamination

Cell line was tested for mycoplasma contamination, and no cell lines were detected positive for mycoplasma contamination via a PCR-based method.

Commonly misidentified lines (See [ICLAC](#) register)

No commonly misidentified cell lines were used

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals	Mice of the C57BL/6 strain were purchased from Department of Experimental Animal Sciences, Peking University Health Science Center, Beijing, China or Jackson Laboratory, USA. Prkaa1-fl/fl mice were purchased from Jackson Laboratory, and Prkaa2-fl/fl mice were purchased from the GemPharmatech Laboratory. Germ-free C57BL/6J were purchased from GemPharmatech Laboratory. Villin Cre mice were gifted by D. L. Gumucio. All animals involved in our experiments were 8-week-old male mice at the beginning of the study. Mice were housed 3-5 per cage, and maintained under standard laboratory conditions with free access to rodent feed and water in Department of Experimental Animal Science, Peking University Health Science Center, Beijing, China or National Cancer Institute, National Institutes of Health (NIH), USA. Specifically, the light was on from 8 a.m. to 8 p.m., with temperature kept at 21-24 °C, and humidity at 40-70%.
Wild animals	This study did not involve wild animals.
Reporting on sex	The findings apply to only male mice, which was consistent with the sex in our human research participants and previous similar studies (PMID: 20432253; PMID: 19556020; PMID: 34880502).
Field-collected samples	Field-collected samples were not used in this study.
Ethics oversight	All animal experiments complied with the protocols for animal use, treatment, and euthanasia approved by Peking University (permit: LA2020480) and the National Cancer Institute Animal Care and Use Committees of NIH (animal protocols: LM-016 and LM-096).

Note that full information on the approval of the study protocol must also be provided in the manuscript.